

Design and Development of Smart Battery Thermal and Power Management System for PV-Battery- Hydrogen Generation System

Submitted in partial fulfilment of the requirements of the degree of

Master of Technology
Sustainable Energy Generation and Storage Technologies

by

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Approval Sheet

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ABSTRACT

This work presents a Smart Battery Thermal and Power Management System for PV-Battery-Hydrogen Generation System for efficient, safe photovoltaic (PV) battery charging, designed for up to 60 V solar input with a 12 V energy-storage battery, capable of compensating for fast PV power fluctuations in a Hydrogen Generation System. In this paper the system uses a microcontroller-based architecture with precision sensing components (INA219 current and voltage sensor, DS18B20 temperature sensor) and a closed-loop control method to govern the charging process. The system integrates proportional–integral–derivative (PID) control in both the PV battery charging unit and the AEM electrolyzer heating stage to ensure efficient, safe, and stable operation. The PID algorithm dynamically adjusts the charging current & voltage based on solar input and battery state-of-charge, with hysteresis-based protection that disconnects the battery when unsafe conditions occur, such as over-voltage above 12.94 V or overheating above 40.5 °C, and resumes only when values fall below 12.54 V or 38 °C.

Real-time data from DS18B20 temperature sensors monitoring battery and electrolyzer cell temperatures. For the AEM electrolyzer, which must operate between 40 °C and 70 °C for optimal efficiency, a 50 W cartridge heater raises the electrolyte and cell assembly to 40 °C within 5-10 minutes under insulation. Safety mechanisms include hardware over-temperature cut-off or overheating above 70 °C and resumes only when values fall below 65 °C, preventing electrolyzer start until stable temperature is achieved. A fault detection module with a fire/smoke sensor that triggers LCD alerts, GSM (SIM900A)-based SMS notifications, and automatic system isolation during faults, allowing for quick human action. This integrated approach ensures automated cold-start capability, precise temperature regulation, efficient PV power utilization, and robust fault protection, thereby enhancing the operational efficiency of green hydrogen production and contributing to sustainable energy storage solutions for renewable applications that prioritize efficiency, safety, and resilience.

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LIST OF ABBREVIATIONS

Acronym	Expansion
I2C	Inter-Integrated Circuit
SPI	Serial Peripheral Interface
USB	Universal Serial Bus
LIPO	Lithium Polymer Battery
LCD	Liquid Crystal Display
SMS	Short Message Service
GSM	Global System for Mobile Communications
IDE	Integrated Development Environment
SOC	State of Charge
PFET	P-channel Field-Effect Transistor
PID	Proportional-Integral-Derivative
PWM	Pulse Width Modulation
SBT&PMS	Smart Battery Thermal and Power Management System

CHAPTER 1

1.1 INTRODUCTION:

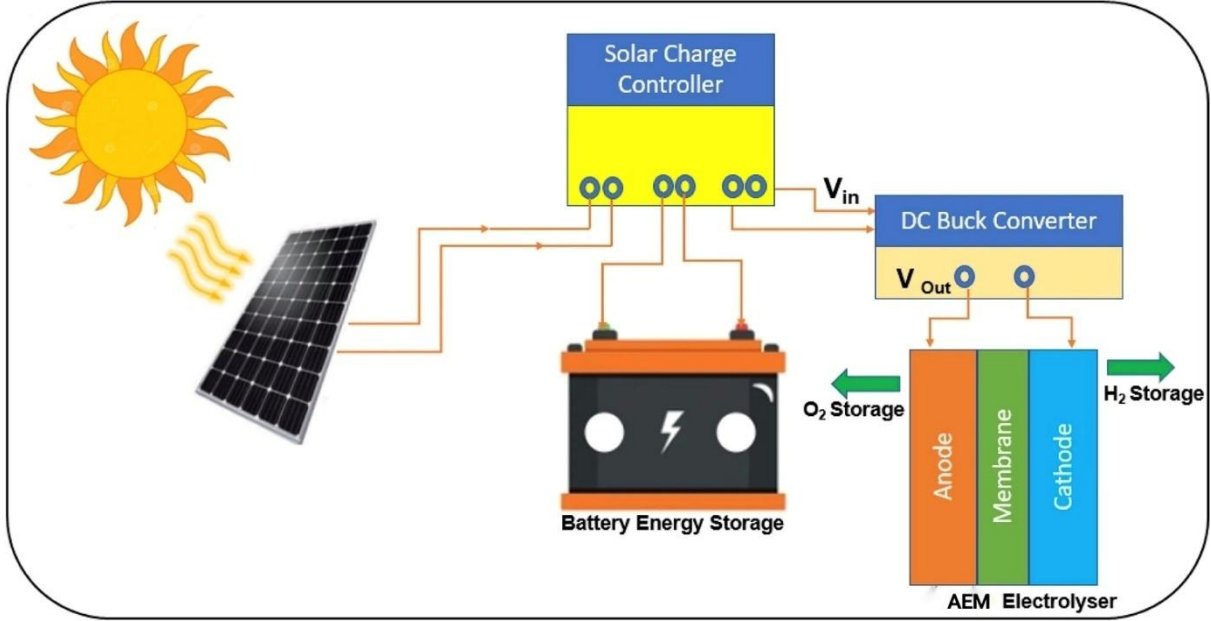


Fig 1.1: Solar Panel, Battery, and AEM Electrolyser with Battery Thermal and Power Management System

The increasing demand for energy, along with the simultaneous need for sustainable and eco-friendly power sources, has driven the adoption of photovoltaic (PV) systems across the globe. In regions lacking centralized electricity infrastructure, solar energy provides a clean, renewable, and cost-effective alternative to traditional fossil fuels. However, the efficiency, safety, and adaptability of solar-powered systems remain a subject of ongoing technical development. One of the most crucial components of any standalone solar system is the power controller responsible for charging the battery bank. These controllers must not only manage the energy transfer between solar panels and storage batteries but must also ensure the system operates safely under diverse environmental and electrical conditions.

The accelerating pace of urbanization, industrial expansion, and population growth has led to a substantial rise in global energy demand. Conventional fossil fuels like coal, oil, and natural gas have historically dominated the global energy landscape; however, their finite availability and significant environmental footprint render them unsustainable for long-term energy security. The combustion of fossil fuels releases large quantities of greenhouse gases (GHGs), primarily carbon dioxide (CO_2), which has intensified global climate change and prompted international commitments toward achieving net-zero emissions by 2050 [1].

Currently, hydrogen production remains largely dependent on fossil-fuel-based routes. In 2021, global hydrogen production reached approximately 94 million tonnes, of which 62% was generated from natural gas via steam methane reforming (SMR) without carbon capture, 19% from coal gasification, and 18% as refinery by-products. These processes collectively emitted more than 900 million tonnes of CO₂ [2]. Although integrating Carbon Capture and Storage (CCS) technologies enables the production of so-called “blue hydrogen,” residual emissions and methane leakage remain environmental concerns. In contrast, water electrolysis powered by renewable electricity offers a truly sustainable pathway for hydrogen production. Despite a 20% growth rate compared to 2020, electrolysis contributed only about 0.04% of global hydrogen supply in 2021 [2], highlighting the need for technological advancements.

The overall thermodynamic potential required for water splitting is 1.23 V under standard conditions. However, in practical systems, additional overpotential (η) is required due to kinetic barriers, particularly for the oxygen evolution reaction (OER), which involves a complex four-electron transfer mechanism. The sluggish kinetics of OER significantly limit overall electrolyzer efficiency and increase operational energy consumption [3].

Table 1.1 Technical comparison of electrolyzers

Characteristics	AWE	AEM	PEM	SOEC
Nominal current density	0.4–1 A/cm ²	0.5–2 A/cm ²	0.2–4 A/cm ²	0.5–1.5 A/cm ²
Voltage range (limits)	1.4–3 V	1.4–2.2 V	1.4–2.5 V	1.0–1.5 V
Stack unit size	1 MW	10 kW – 1 MW (developing scale)	1 MW	5 kW
Electrode area	10,000–30000 cm ²	200–2000 cm ²	1500 cm ²	200 cm ²
Cold start-up (to nominal load)	30–60 min	<5 min	<5 min	>600 min
Warm start-up (to nominal load)	1–5 min	<1 min	<10 s	10–15 min
Load range	15%–100 %	10%–120%	5%–120 %	30%–125 %
Ramp rate	0.17 %/s–0.33 %/s	5–10 %/s	>10 %/s	≈0.083 %/s
Efficiency	68–77%	65–75% (LHV basis)	70–80%	80–90%

Alkaline electrolyser (AEL), PEM electrolyser, solid oxide electrolyser (SOEL), and anion exchange membrane (AEM) electrolyser are the main types of electrolyzers .

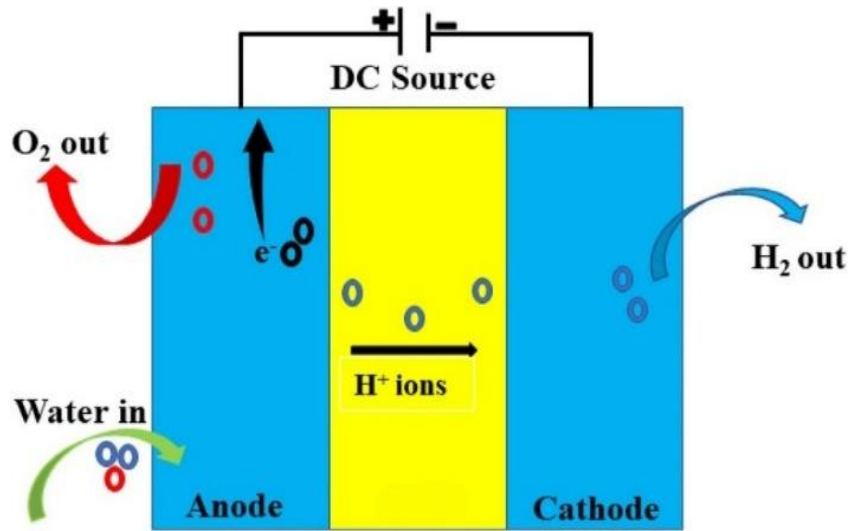


Fig 1.2: AEM electrolyser Basic Working Principle

The AEM electrolyser has several advantages, such as an extensive operating temperature range (40–70 °C), high current density, reduced gas permeability, enhanced energy efficiency, and ease of handling and maintenance.

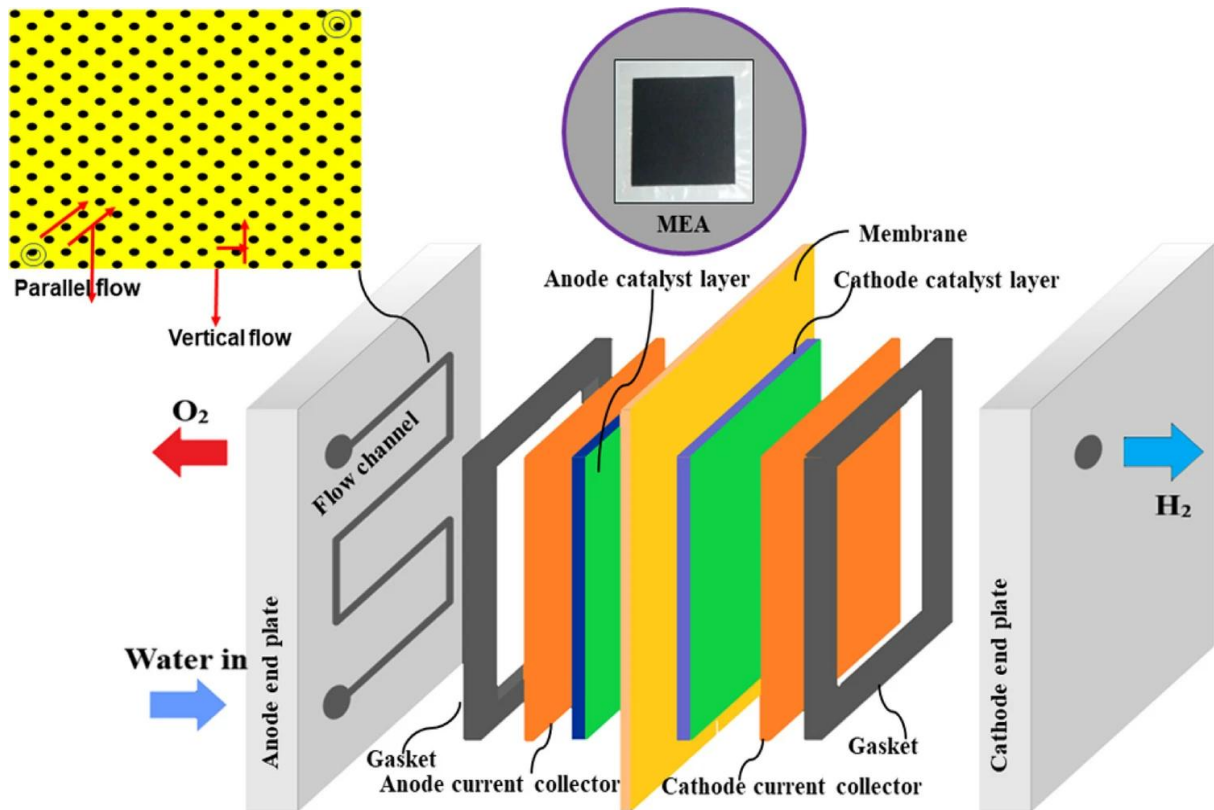


Fig 1.3: AEM electrolyser Internal Core Construction

1.2 Background of the Problem

The increasing demand for clean and sustainable energy has accelerated the adoption of photovoltaic (PV) systems integrated with energy storage and hydrogen production technologies such as Anion Exchange Membrane (AEM) electrolyzers. However, the effective operation of a PV–Battery–AEM electrolyser system remains a significant challenge due to the intermittent and highly variable nature of solar energy.

Conventional photovoltaic systems typically rely on basic charge controllers that follow fixed and predefined control algorithms for battery charging. These controllers lack the capability to adapt dynamically to changing environmental conditions such as solar irradiance, temperature variations, and fluctuating load demands. As a result, inefficient energy utilization, battery degradation, and suboptimal hydrogen production can occur.

In hybrid systems involving batteries and AEM electrolyzers, the problem becomes more complex. The battery must be charged efficiently while simultaneously supplying stable and controlled power to the electrolyser. However, variations in PV output can lead to unstable charging currents, voltage fluctuations, and thermal stress within the battery. These factors not only reduce battery lifespan but also affect the performance and efficiency of the electrolyser, which requires a consistent power input for optimal hydrogen generation.

Moreover, thermal management is a critical issue in such systems. Batteries are highly sensitive to temperature, and excessive heat can lead to safety risks, reduced efficiency, and permanent damage. Similarly, improper voltage regulation can result in overcharging or deep discharge conditions, further degrading system reliability. Traditional systems often lack integrated mechanisms for real-time monitoring and protection against over-temperature and over-voltage conditions. Another key limitation is the absence of intelligent fault detection and adaptive control strategies. Without continuous monitoring of parameters such as current, voltage, and temperature, it becomes difficult to detect abnormal operating conditions and take corrective actions in real time. This can lead to system inefficiencies, unexpected shutdowns, and potential damage to critical components. Therefore, there is a clear need for an advanced, intelligent power management system capable of dynamically controlling energy flow between the PV panel, battery, and AEM electrolyser. Such a system should incorporate real-time sensing, adaptive control algorithms, and robust safety mechanisms to ensure efficient energy utilization, enhanced battery life, and stable hydrogen production.

1.3 Research Motivation and Objective

Existing PV-hydrogen systems lack fast fluctuation handling in hardware level.

Multi-subsystem integrated control architecture has not yet configured to optimize coordinated energy distribution between the photovoltaic unit, battery subsystem, and electrolyzer based on real-time system conditions. In addition, Automated cold-start, precise thermal control, Hysteresis protection, Automatic fault-triggered isolation mechanism and fault protection haven't been implemented.

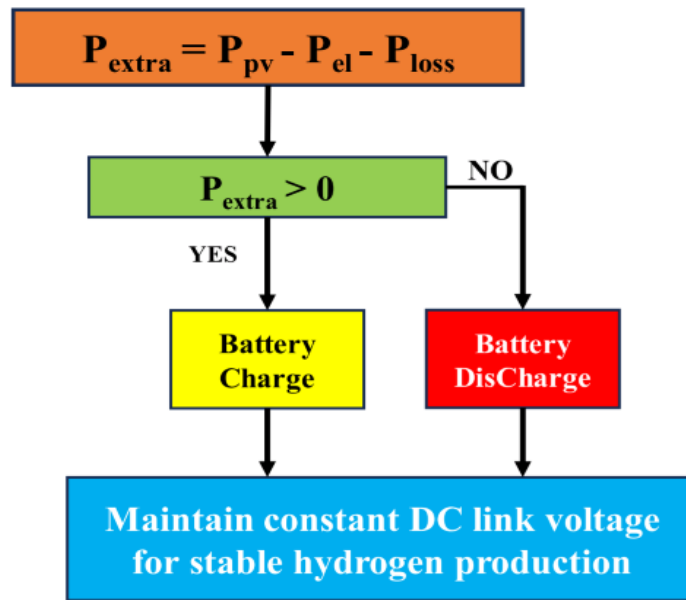


Fig 1.4: Block Diagram representing the System Design & Power Flow

Therefore, this work addresses the identified research gaps through the following Contributions:

1. The multi-subsystem integrated control architecture is configured to optimize coordinated energy distribution between the photovoltaic unit, battery subsystem, and electrolyzer based on real-time system conditions.
2. A real-time monitoring module logic configured continuously to acquires voltage, current, and temperature data from multiple sensing nodes distributed across the subsystems.
3. Hysteresis-based protection logic configured to prevent instability caused by rapid electrical and thermal fluctuations.
4. Automatic fault-triggered isolation mechanism including a lockout functionality configured to isolate at least one subsystem upon detection of abnormal operating conditions and to permit resumption of operation only after restoration of stable conditions.
5. Electrolyzer control circuit configured to enable automatic cold-start operation and regulate temperature during initial startup conditions.

CHAPTER 2

Literature Review

The increasing worldwide energy consumption and the need for sustainable solutions have accelerated PV system adoption. Renewable energy like Solar Photo Voltaic Cells offers a clean and economical alternative, but standalone PV systems still require improvements in efficiency, safety, and adaptive control. Where particularly in power controllers that manage PV-to-battery charging under varying environmental and electrical conditions. Research on PV-Battery-Electrolyzer systems began with Kikuchi et al. [1], who showed that harmonizing PV output with battery storage enhances economic feasibility, reduces electrolyzer oversizing, and stabilizes the DC link during off-grid fluctuations, establishing battery-assisted smoothing as essential.

Later, Xin Cai et al. [2] demonstrated that real PV fluctuations accelerate PEM degradation, doubling voltage attenuation (7.8 mV/h at 1 A/cm²) and causing an irreversible 48.3% MEA reaction-area loss, confirming the need for stable power conditioning with advanced sensing and protective circuitry. Parallel research addressed PV–electrolyzer mismatch and power-processing challenges in [3] introduced an IMPPT neural-network method achieving 91% conversion efficiency than conventional P&O by advancing intelligent PV–electrolyzer coordination. Yodwong et al. [4] showed that thyristor rectifiers generate ripple and reactive-power issues that increase specific energy consumption, recommending high-frequency DC/DC converters, interleaving, and active filtering. Alonge et al. [5] proposed an H₂/H_∞ controller for isolated converters in micro-wind electrolyzer systems. Şahin [6] demonstrated via MATLAB/Simulink that coordinated MPPT and buck-converter regulation achieve >90% efficiency with stable voltage/current for reliable hydrogen production.

Awad et al. [7] highlighted that maintaining PEM electrolyzers within 50–80 °C is essential for membrane life. For each 10 °C rise beyond this window halves membrane durability due to increased ohmic loss, catalyst corrosion, and non-uniform current distribution. Their recommendations advanced cooling, PCM integration, and strengthened membranes form the foundation for modern multi-sensor thermal monitoring. Battery-buffered stabilization was experimentally validated by Kumar and Samuel [8], Al-Mut et al. [9] introduced a hybrid PSO–PID MPPT that improved convergence and reduced tracking errors, indirectly protecting electrolyzers from thermal and electrical stress.

Reliability advances were strengthened by Taghezouit et al. [10], whose model-based and ML-assisted diagnostics one-diode modelling, SVR, and PCA anomaly detection enabled predictive fault identification. Diagnostic precision was further elevated by Dhanraj et al., who integrated digital signal processing with machine learning classifiers to transform raw thermal and electrical data into actionable fault identification [11]. Suresh Muthusamy pioneered real-time operational awareness in solar energy by leveraging IoT architecture and cloud integration to enable continuous remote monitoring and automated performance maintenance [12]. Addressing the challenges of intermittent solar power, Peter Makeen & Hani introduced an intelligent charging controller using artificial neural networks and LSTM forecasting to stabilize off-board electric vehicle charging [13]. Wang and Nehrir pioneered a coordinated power management strategy for hybrid wind/photovoltaic systems, utilizing fuel cell-electrolyzer stacks to effectively reduce the difference between fluctuating energy production and continuous load demand [14]. Valenciaga and Puleston established a robust supervisor control architecture for hybrid systems, utilizing non-linear control techniques to balance the erratic nature of wind and solar energy while maintaining optimal battery health [15].

Senjyu et al. advanced operational resilience in remote locations by integrating frequency control and battery energy storage for stable, autonomous power in island grids using wind and solar energy [16]. Li et al. established a solid oxide fuel cell (SOFC) load-following control strategy that stabilizes isolated grids during power imbalances by mitigating frequency fluctuations [17]. Agbossou et al. demonstrate that a standalone, renewable integration utilizing hydrogen energy carrier and fuel cell technology effectively bridges seasonal gaps in solar and wind power generation [18]. Abdelghany et al. (2024) proposed an integrated, optimized control approach for grid-tied hybrid wind-solar microgrids to mitigate power fluctuations and ensure grid integration [19]. Abomazid et al. developed a stochastic energy management framework to optimize the integration of hydrogen facilities with battery storage to mitigate solar PV intermittency [20]. A new multi-layered optimization framework developed by Ali et al. incorporates Hydrogen-based Technologies & energy load management to strengthen the integration of EV's & Sustainable source across Energy source networks [21].

CHAPTER 3

Design and methodology

The below **Fig 3.1** how the multi-subsystem integrated control architecture the configuration helps to optimize coordinated energy distribution between the photovoltaic unit, battery subsystem, and electrolyzer based on real-time system conditions.

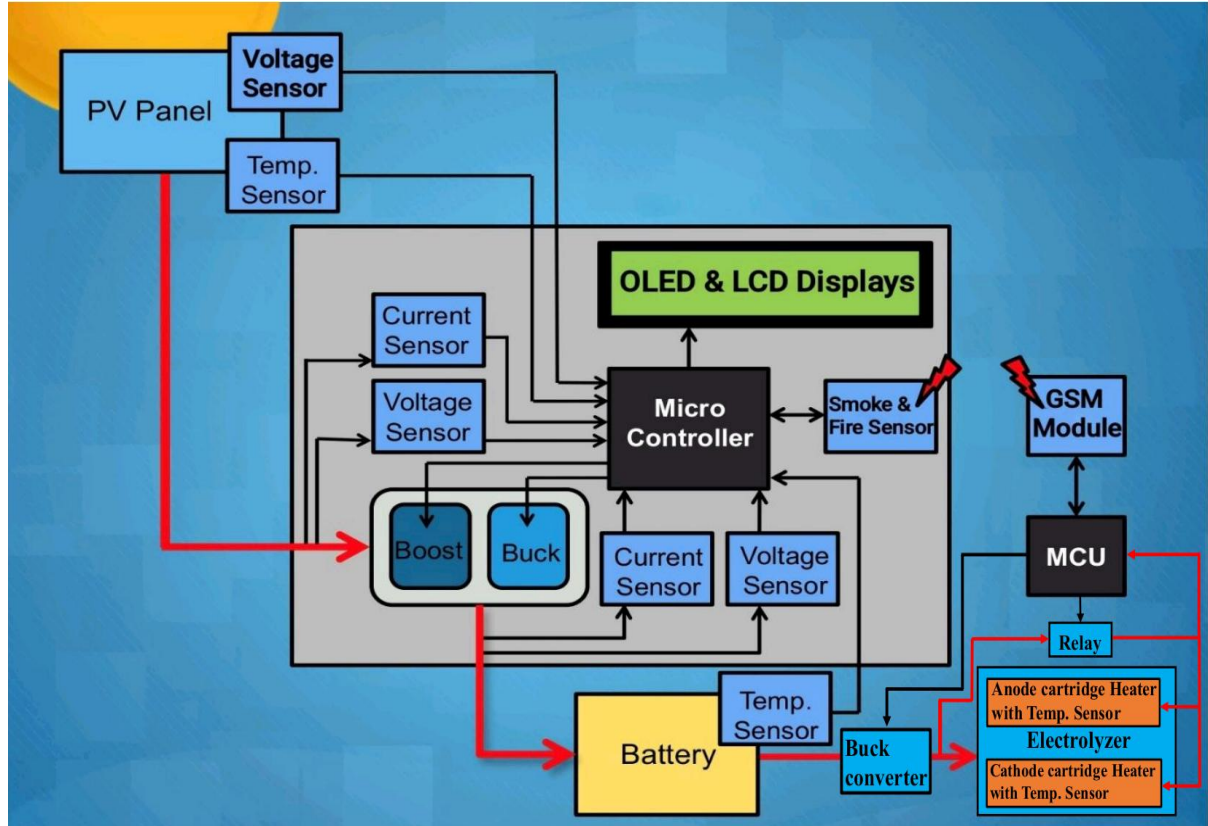


Fig 3.1: The multi-subsystem integrated control architecture

3.1. Components Abbreviations:

Table 3.1: Components Abbreviations

C : Capacitor	MOSFET : Metal Oxide Semiconductor Field Effect Transistor
BESS : Battery energy storage systems	PEMEL : Polymer Electrolyte Membrane Electrolyzer
D : Duty cycle	F : Farad
P : Power	IRENA : International Renewable Energy Agency
HHV : Higher heating value	F_{H2} : Electrolyzer hydrogen flow rate (mol/s)
AEM : Anion Exchange Membrane	PV : Photovoltaic
ST : Storage tank	NSGA-II : Non dominated Sorting Genetic Algorithm - II
PMS : Power management system	N : Electrolyzer cell number
KPI : Key performance indicators	HRES : Hybrid renewable energy system
AEL : Alkaline electrolyzer	°C : Degrees Celsius
SPE : Solid polymer electrolyte	GSM : Global System for Mobile Communications
ηF : Faraday efficiency	SMS : Short Message Service.
L : Inductor	C_p : The power coefficient of wind turbine
SIM : Subscriber Identity Module	NREL : National renewable energy laboratory
p.u : Per unit	η : Number of electron exchanges
P&O : Perturb & observe	H : Henry
SMS : Short Message Service.	PEM : Polymer electrolyte membrane
LCOE : Levelized cost of electricity	

3.2. Anion Exchange Membrane electrolyzer

The AEM electrolyzer operates in an alkaline environment, often with a low-concentration alkaline solution like potassium hydroxide (KOH) as an electrolyte to enhance conductivity.

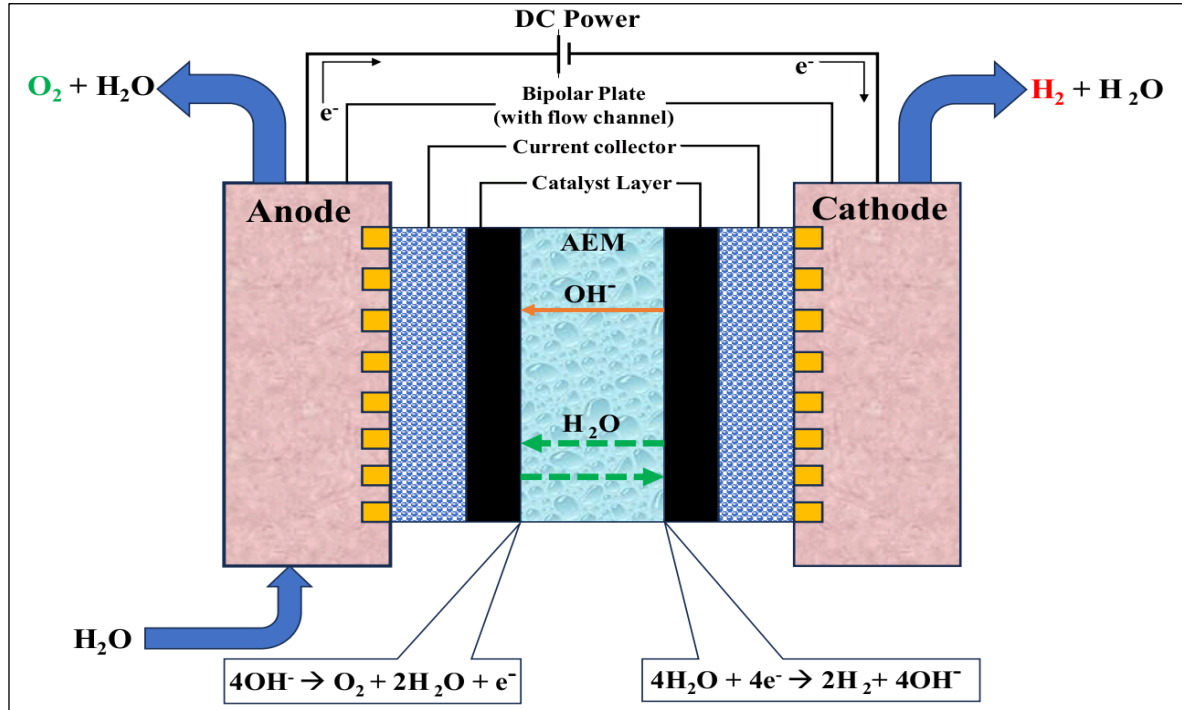


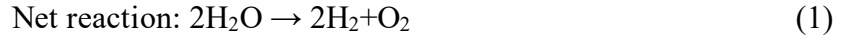
Fig 3.2: AEM electrolyser Basic Working Principle

Table 3.2: AEM Cell parameters

Voltage range	1.7-3v(max)
Current Density	0.5A/cm ² to 2 A/cm ²
Cell Fixture	AEM electrolyser
Model	TES-CF-AEM-9-M2
Number of Cells	Single cell
Membrane	Standard -Anion Exchange membrane
Electrolyte	Potassium Hydroxide (KOH) @1M
Flow Configuration	Single side / Dual side
Surface Area	3x3 cm = 9cm ²
Temperature Range	40°C to 80°C
Cell Torque	6 - 9 Nm (* Based on Loading Conditions)
Cell Weight	1.2 kg (*Loading Conditions)
Cell Dimensions (LXB)	100 mm x 100 mm
Materials of Construction	Stainless steel (SS316)
Sealing Mechanism	Gasket
Leak-Less Cell Design	Yes

The Main Advantage over other Electrolyzers employ cost-effective metal catalysts, such as nickel- and iron-based compounds instead of costly platinum-group metals required in PEM electrolyzers and its core reaction is slightly acidic in nature.

The chemical reactions involved in AEM water electrolysis are as follows:



The hydrogen generation rate of the AEMEL can be determined using the equation below [8]:

$$F_{\text{H}_2} = \frac{\eta I}{2F} \quad (2)$$

Faraday's efficiency is expressed as using empirical Faradaic efficiency correlation:[7]

$$F_{\text{H}_2} = 96.5 \exp\left(\frac{0.09}{i_{el}} - \frac{75.5}{i_{el}}\right) \quad (3)$$

Where, (electrolyzer current density) $i_{el} = \frac{I}{A}$

The hydrogen generation rate, expressed in Nm³/h, is calculated as follows:

$$Q = F_{\text{H}_2} \times 3600 \times 0.022414 \quad (4)$$

3.3. Experimental Requirements and workable ranges for all the parameters

Table 3.3 Experimental Requirements and workable ranges for all the parameters

S.No.	Sub-component	Specification
1	Solar PV Module	24V, 60W, Mono-crystalline
2	LiPo Battery	11.1V, 2200mah, 30C
3	DC-DC Buck Converter Module (LM2596)	12V to 5V
4	P-Channel MOSFET (Q1 – IRF9540N)	100V, 19A
5	NPN Transistor (Q2 – 2N2222)	40V, 600mA
6	Schottky Diode (D1 – 1N5822)	40V, 5A
7	1N4007 Diode	±1% (typical)
8	Inductor	330 μH,
9	Capacitors	100 μF, 25V
10	5V Single-Channel Relay Module	50VAC or 30VDC, 10A
11	Micro Controller (Arduino Nano)	ATmega328P, 5V, 200mA, 32KB, 16MHZ
12	INA219 Current Sensor	5V, 5A
13	DS18B20 Digital temperature sensor	5V, 750 ms (max), -55°C to +125°C (or -67°F to +257°F)
14	MQ-4 Smoke Sensor Specifications	300~10000ppm, 5V, 40mA
15	GSM Module (SIM900A)	12V, 1.5mA
16	OLED Display (128x64)	5V, 10mA, 9-bit format
17	16x2 LCD Display	5V, 5mA
18	Resistors	4.7kΩ, 2kΩ
19	Cartridge Heater –	230V(VAC), 100V(VDC), 50W, 150°C
20	AEM Electrolyzer	1.7-3v(max), 0.5A/cm ² to 2 A/cm ² , 40°C to 80°C

CHAPTER 4

DESIGN & IMPLEMENTATION FRAMEWORK

4.1 Overall simple Circuit Diagram of Smart Battery Thermal and Power Management System for PV-Battery-Hydrogen Generation System

The proposed system shown in Fig. 4.1 is an integrated solution designed to efficiently manage solar energy, monitor critical parameters, and ensure system safety. It uses a microcontroller to collect real-time data from voltage, current, temperature, and smoke/fire sensors connected to the PV panel, battery, and load. Logically the overall system is divided into **three major sub block circuits** based on functionality, signal flow, and control objectives, which is particularly important for hybrid PV–battery–electrolyzer systems.

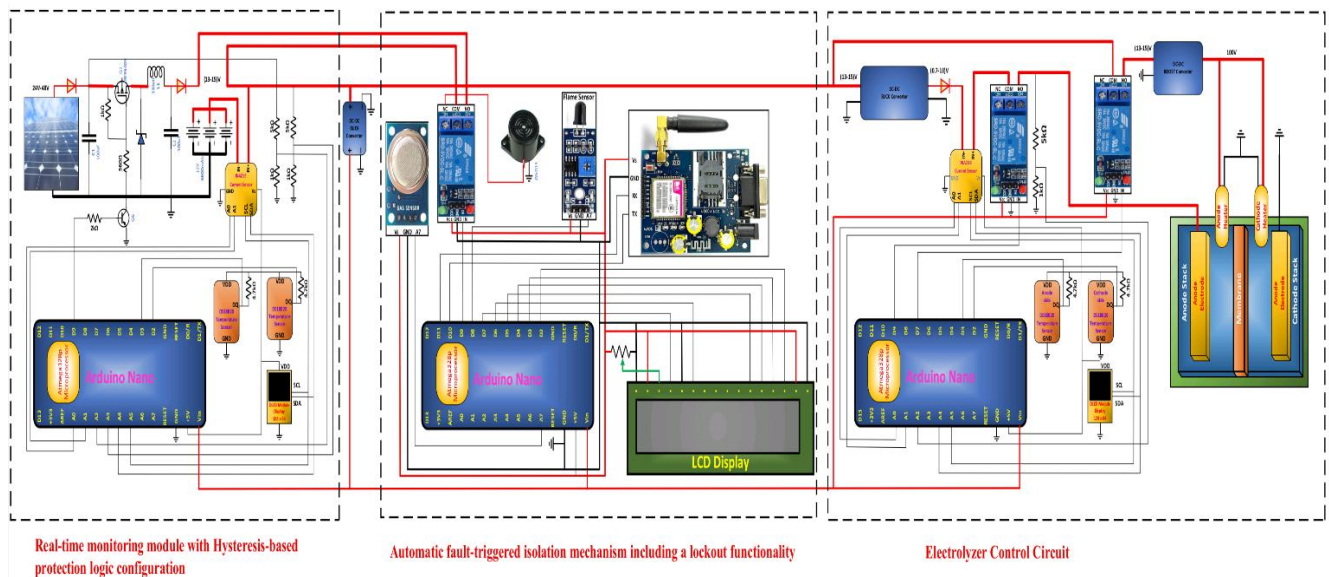


Fig 4.1: Overall simple Circuit Diagram of Smart Battery Thermal and Power Management System for PV-Battery-Hydrogen Generation System

Major Objectives –

I: Smart Battery Thermal and Power Management System (Fig 4.4)

For real-time monitoring module configuration with Hysteresis-based protection logic which continuously acquires voltage, current & temperature data from multiple sensing nodes distributed across the subsystems.

II: Circuit Fault Detection and Alert System (Fig 4.7)

For Automatic fault-triggered isolation mechanism including a lockout functionality to isolate at least one subsystem upon detection of abnormal operating conditions and to permit resumption of operation only after restoration of stable conditions and Alert System through SMS.

III: Electrolyzer Control Circuit (Fig 4.10)

To enable automatic cold-start operation and regulate temperature during initial startup conditions.

4.3 Equivalent Circuit Model of Smart Battery Thermal & Power Management System I:

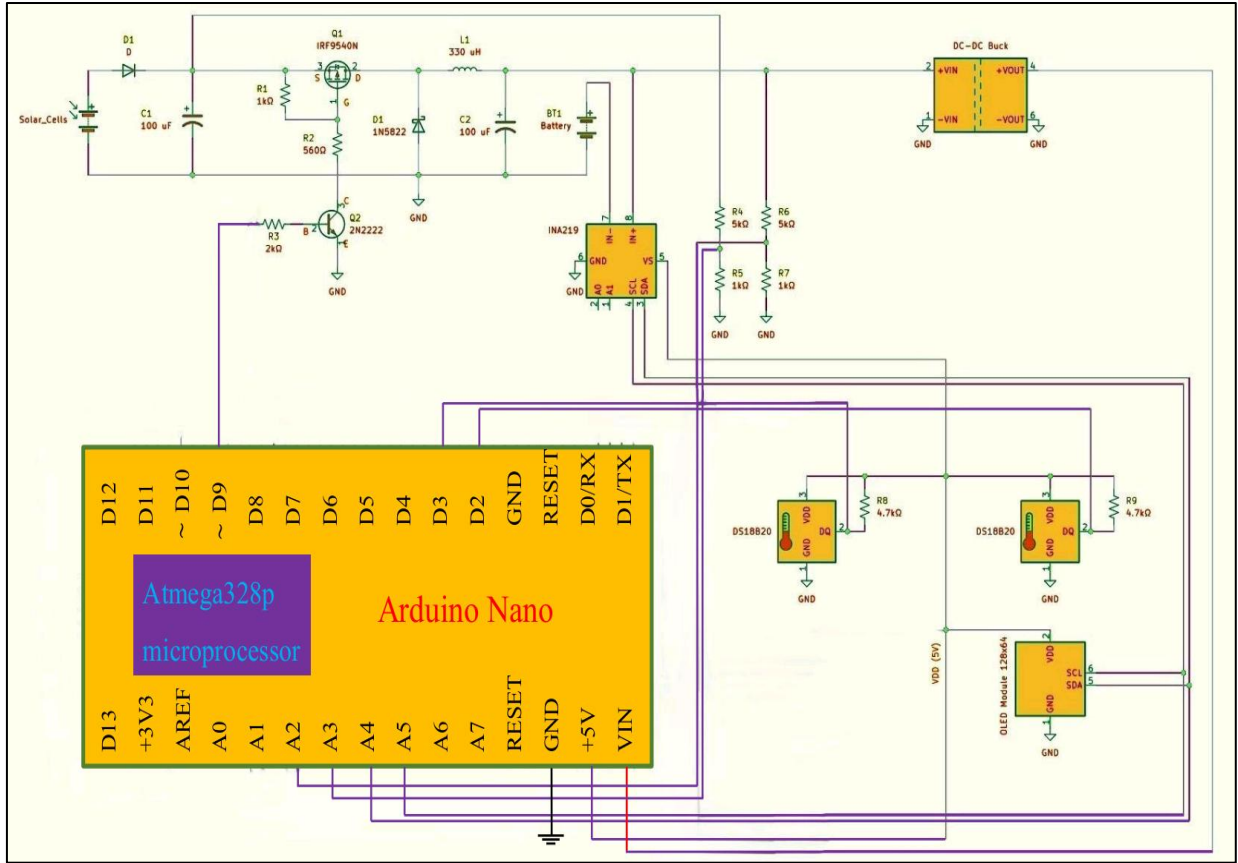


Fig 4.4: Equivalent Circuit Model of Smart Battery Thermal and Power Management System

The equivalent circuit model of the Smart Battery Thermal and Power Management System design circuit is shown above this charging system that intelligently manages energy from a photovoltaic (PV) panel to charge a battery safely and efficiently. At the hardware level, the PV panel generates DC electricity, which is first sensed for voltage and current using a voltage divider and the INA219 current sensor module. The sensed parameters are fed to an Arduino Mega, which acts as the system's microcontroller. To condition this raw solar energy, a boost converter circuit comprising a power MOSFET (Q1), inductor (L1), and diode (D1) is controlled via PWM from the Arduino Nano (pin 9). This regulation ensures that adequate voltage is supplied for charging, especially under variable sunlight conditions. The regulated output is directed to charge a 12V battery, with a buck converter added for stepping down voltage for other loads or devices when needed.

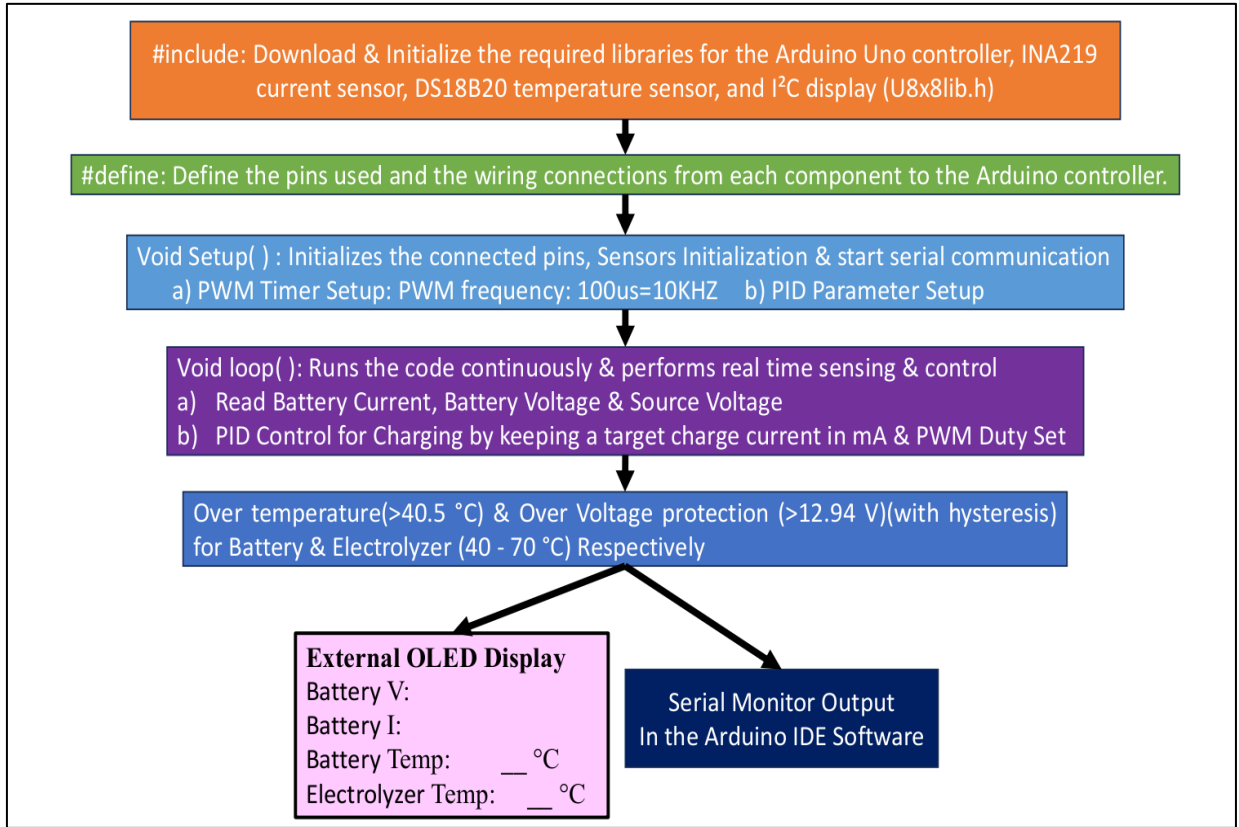


Fig 4.5: Arduino Code flowchart for PV Battery Charging with PID Control

The Arduino gathers critical real-time data using additional sensors: two DS18B20 temperature sensors monitor the battery temperature and ambient conditions respectively, while additional voltage sensors track both source (solar) and battery voltages. These inputs feed into a PID-based control algorithm coded inside the Arduino, ensuring dynamic adjustment of the PWM signal that regulates charging current. If conditions become unsafe such as over-voltage (above 12.94V) or overheating (above 40.5°C) the code halts PWM to stop charging and only resumes once safe thresholds are re-entered (e.g., below 12.54V or 38°C). This creates hysteresis logic that prevents rapid switching and enhances safety during borderline conditions.

The software implementation of the control system includes a custom Simple PID class that calculates an optimal PWM duty cycle based on errors between target and actual voltage/current. This allows smoother control of charging current and avoids battery stress or excessive ripple. The system also calculates the state-of-charge (SoC) percentage based on battery voltage and current data, helping modulate the power delivery more gently as the battery nears full charge. In addition to control, the code logs real-time values and acts as a decision-maker to balance performance with battery health, all while working autonomously without requiring user intervention.

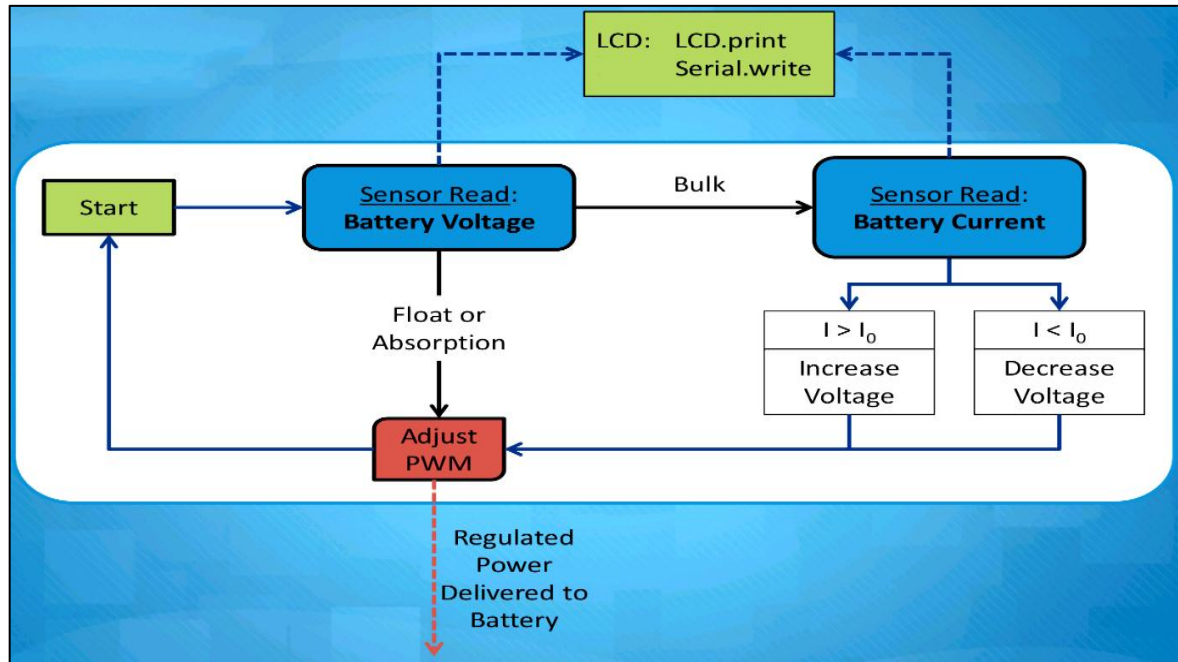


Fig 4.6: PWM-Based Battery Charging Algorithm with Voltage and Current Feedback

Table 4.1: Pin Allocation of Arduino Nano for Components and Sensors

Component or Sensors	pin	Function
PFET PWM Control	D9	Uses Timer1. PWM (). Safe because Timer1 is dedicated and not shared with other libraries. This pin controls the PFET (Power MOSFET) gate for battery charging regulation. Keeping it isolated avoids timing conflicts.
Battery Temp Sensor	D2	Measures battery temperature for safety. Initially conflicted with interrupts but works fine now. Connected via One Wire bus (Dallas Temperature /DS18B20 sensor).
Electrolyzer Temp Sensor	D3	Electrolyzer temperature allows independent monitoring and safety-based control (e.g., power shutoff if Electrolyzer temp too high).
I2C OLED Display (U8x8)	A4 & A5	Standard I2C pins: SDA (A4), SCL (A5). Dedicated for OLED display communication. Important: keep reserved for I2C only, don't reassign.
Solar Voltage Input	A3	Analog input pin for sensing voltage (e.g., solar input voltage). Connects to a voltage divider to scale down input to safe Arduino levels (0–5 V).
Battery Voltage Input	A2	Another analog input pin, used for battery voltage measurement. Also goes through a voltage divider for safe measurement.

4.4. Equivalent Circuit Model of the Circuit Fault Detection and Alert System -II:

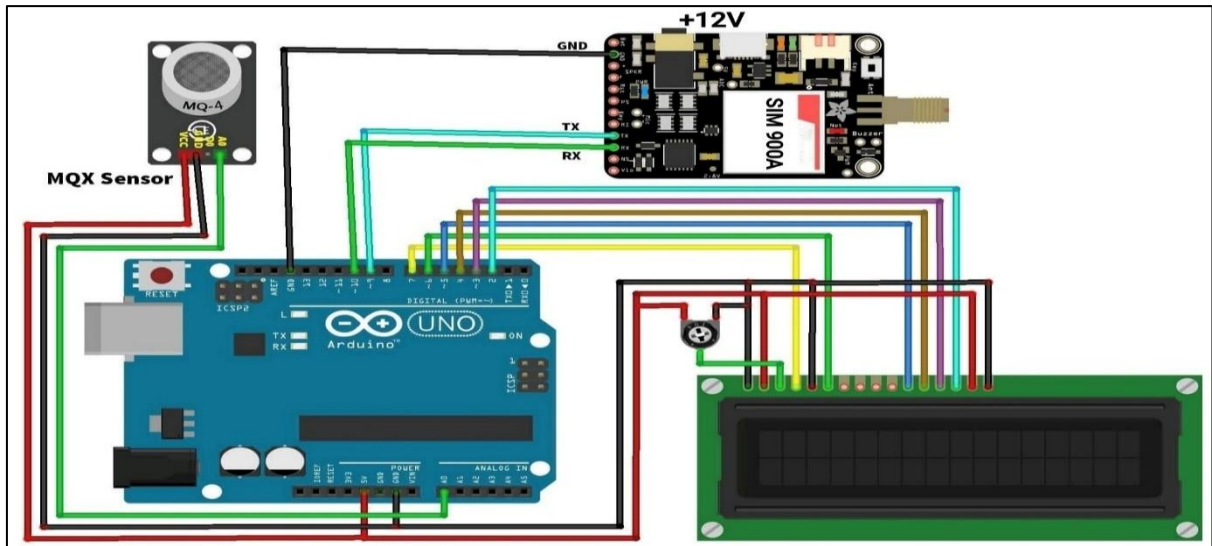


Fig 4.7: Equivalent Circuit Model of the Circuit Fault Detection and Alert System

The above equivalent circuit model of the Circuit Fault Detection and Alert System using an Arduino Uno, MQ-4 gas sensor, GSM SIM900A module, and a 16x2 LCD display. The MQ-4 sensor detects combustible gases like methane (CH_4) and outputs an analog signal to the Arduino's A0 pin. The Arduino continuously monitors this signal, and if the gas concentration exceeds a certain threshold (value > 500), it considers it a fault condition. The LCD, connected to digital pins 7 through 2, displays real-time status messages such as the current gas level or a warning if a fault is detected. This ensures the user has immediate visual feedback about the circuit or environmental condition. The LCD first shows a welcome message on startup and then updates in real-time during loop execution based on the gas sensor readings.

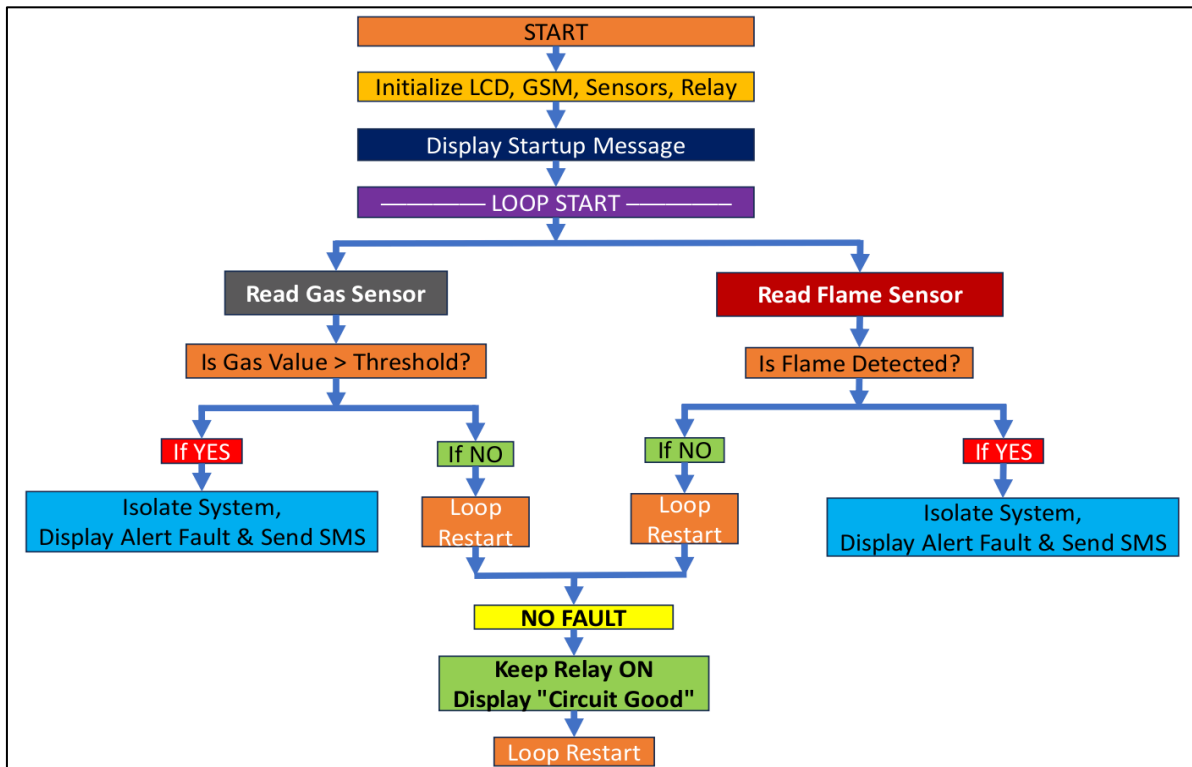


Fig 4.8: Arduino Code flowchart for the Circuit Fault Detection and Alert System

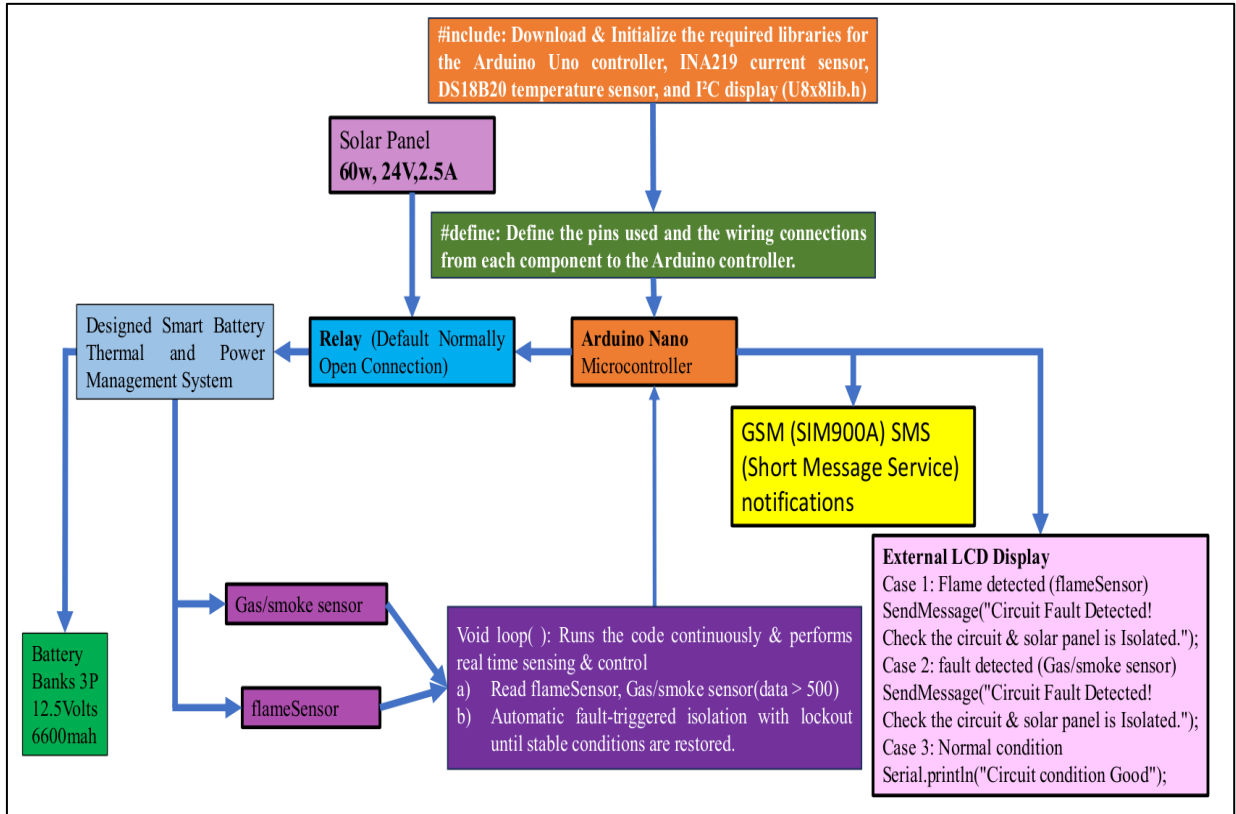


Fig 4.9: Equivalent Block Diagram Model of the Circuit Fault Detection and Alert System

A standout feature of this system is its remote alert functionality using the SIM900A GSM module, which is connected via software serial (pins 10 for Rx and 11 for Tx). When a fault is detected, the Arduino sends AT commands through the Send Message () function to initialize the module, switch it to text mode, and transmit a predefined SMS alert to a specified phone number. This allows real-time alerts to be sent without relying on internet connectivity. The SIM900A is powered by a separate 12V supply to meet its higher current requirements. The serial monitor also logs live readings and fault messages for debugging or monitoring via a computer. This integration of real-time sensing, display, and GSM communication makes the system highly suitable for applications in industrial safety, electrical labs, and residential gas leak monitoring where both immediate local notification and remote alerts are critical.

4.5. Equivalent Circuit Model of the Electrolyzer Control Circuit - III:

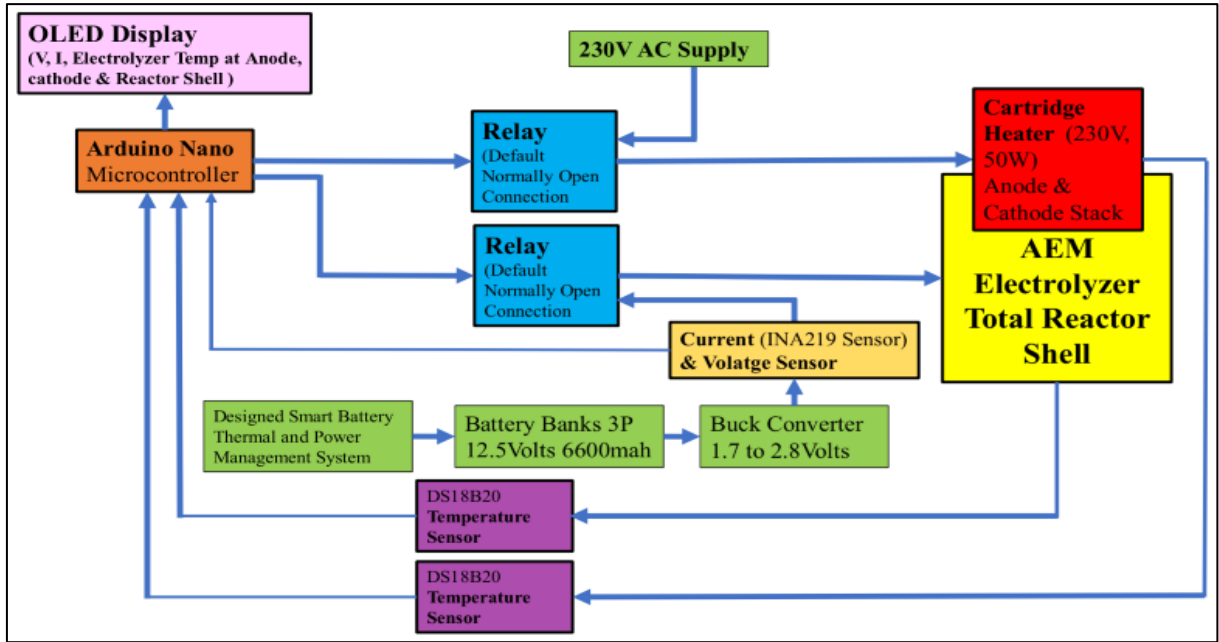


Fig 4.10: Electrolyzer Control Circuit

The **Electrolyzer Control Circuit** system developed with the integration of battery-driven DC supply, a controlled AC heating unit (either On or Off), and a complete sensing and monitoring arrangement to operate an AEM electrolyzer under regulated conditions. A 12.5-V, 6600-mAh (3P) battery bank serves as the primary source, protected by a dedicated Smart Battery Thermal and Power Management System that handles cell balancing, temperature supervision, over- and under-voltage protection, and current-limit safeguards before enabling the pack's output. The battery feeds a buck converter, which is adjusted to deliver the required 1.7–2.8 V operating range for the electrolyzer. This output is routed through an INA219 (current sensor) high-side current and voltage monitor, allowing accurate measurement of both the stack voltage and reaction current. All DC connections including those of the buck stage, the INA219 (current sensor), and the Arduino Nano, are kept common to maintain stable referencing. The Arduino, powered by a regulated 5-V rail, manages the sensors, display, and control logic.

The electrolyzer stack receives the regulated low-voltage (1.7-2.8v) DC output, while three DS18B20 digital temperature sensors placed at the anode, cathode, and reactor shell provide point-wise thermal monitoring over a single one-wire data line with a standard pull-up.

An OLED module connected on the I²C bus displays real-time temperature, voltage, and current values. To maintain the reactor at its intended operating temperature, a 230-V AC, 50-W cartridge heater is fitted to the reactor body and switched through relays whose default condition is normally open. These relays are controlled by the Arduino and may be substituted with an appropriately rated SSR if smoother AC control or zero-cross switching is required.

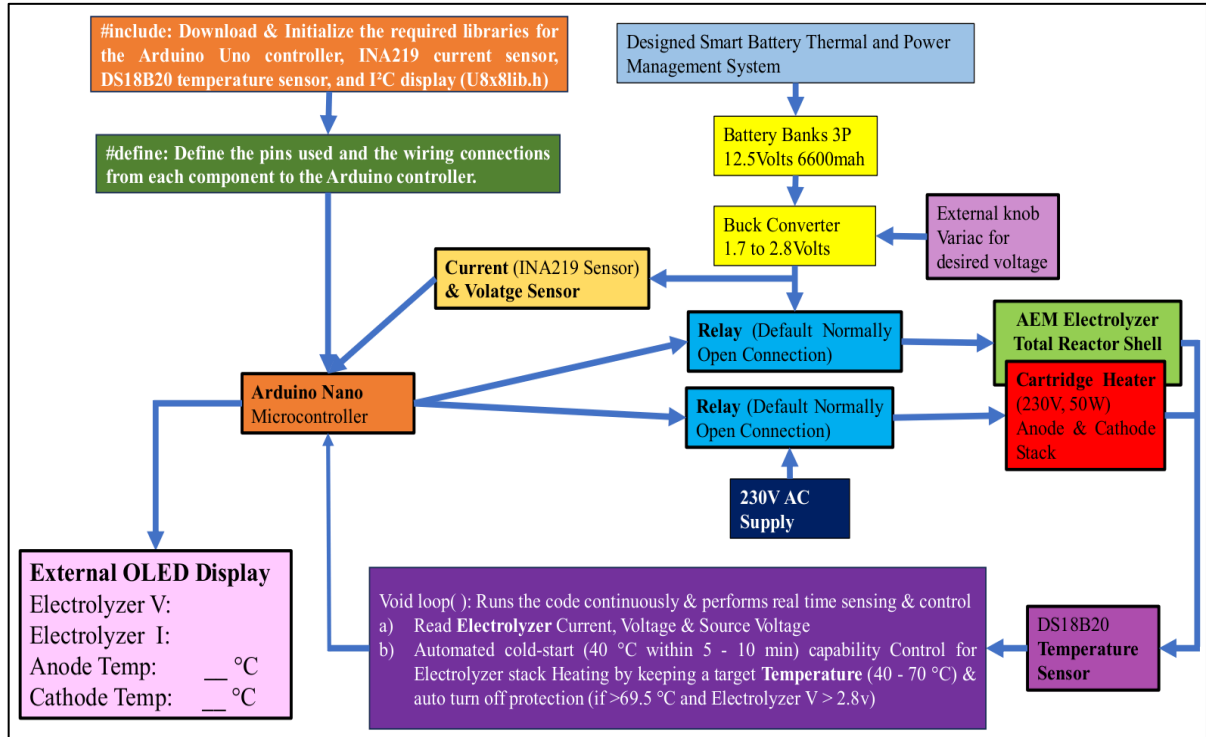


Fig 4.11: Arduino Code flowchart for the Electrolyzer control circuit

The mains wiring for the heater routes the live conductor through the relay contacts, while neutral and earth are handled according to electrical-safety practice, supported by suitable fusing and RCD/GFCI protection. The Arduino reads thermal data continuously and applies either hysteresis-based or PID-based decisions to control heater operation, while also enforcing safety cutoffs in the event of abnormal temperature rise or sensor failure. During startup, the BMS verifies battery integrity before enabling the DC line, after which the Arduino nano (micro controller) initializes the INA219 (Current sensor), detects all DS18B20(temp sensor) devices on the one-wire bus, and confirms by displaying the operation condition of the **Electrolyzer Control Circuit** data in OLED Display. The buck converter is then set to the intended or required electrolyzer voltage, and the system begins regular monitoring of reaction current, stack voltage, and all three temperature points. The heater is energized only when necessary to achieve the target operating temperature and is immediately shut off if any reading exceeds predefined limits. The arrangement ensures accurate tracking of the electrolyzer's electrical behaviour while maintaining strict thermal control. Additional safety comes from hydrogen venting considerations, proper segregation of mains and low-voltage wiring, common grounding on the DC side, and fuses placed both on the battery and the AC heater line. With these elements combined, the circuit provides a complete, self-contained platform for controlled AEM electrolyzer operation, featuring integrated measurement, thermal regulation, battery protection, and secure switching of the heater load.

CHAPTER 5

Results and Discussion

5.1 Developed Smart Battery Thermal and Power Management System for PV-Battery-Hydrogen Generation System:

To evaluate the Electrolyzer behaviour, battery behaviour, thermal response, power-regulation accuracy, and protection performance of the developed Smart Battery Thermal and Power Management System for the PV-Battery-Hydrogen Generation System, a five-hour controlled test was conducted in which every critical parameter including battery voltage, charging and load current, multi-point temperature readings, buck-converter output, and the system's reaction to rapid PV-power fluctuations was continuously recorded.

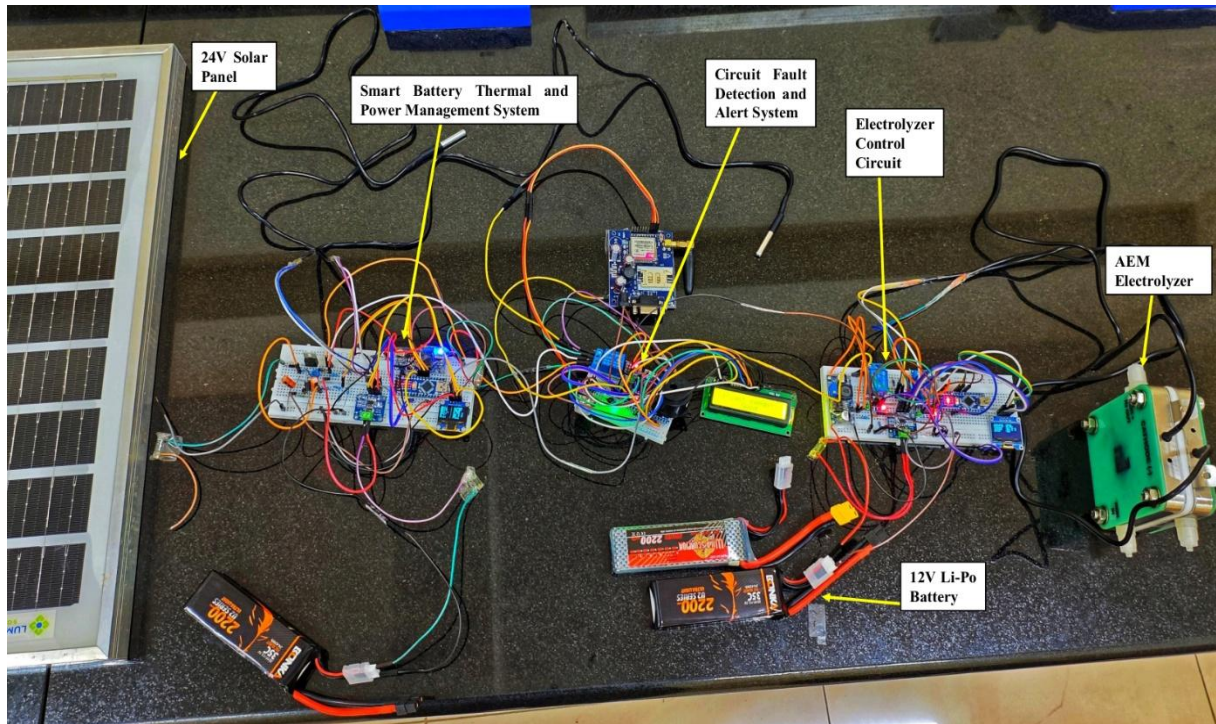


Fig 5.1: Developed Smart Battery Thermal and Power Management System for PV-Battery-Hydrogen Generation System

The collected figures and tables summarize these results, capturing trends in voltage progression, stability of the control algorithm, and the behaviour of the thermal-management features under varying load and ambient conditions. All transient events such as current spikes, voltage dips, and temperature deviations were also logged, providing complete experimental evidence of the system's operational stability, charging accuracy, and integrated protection capability during operation.

Time-Domain Output Voltage Response from MATLAB Simulation:

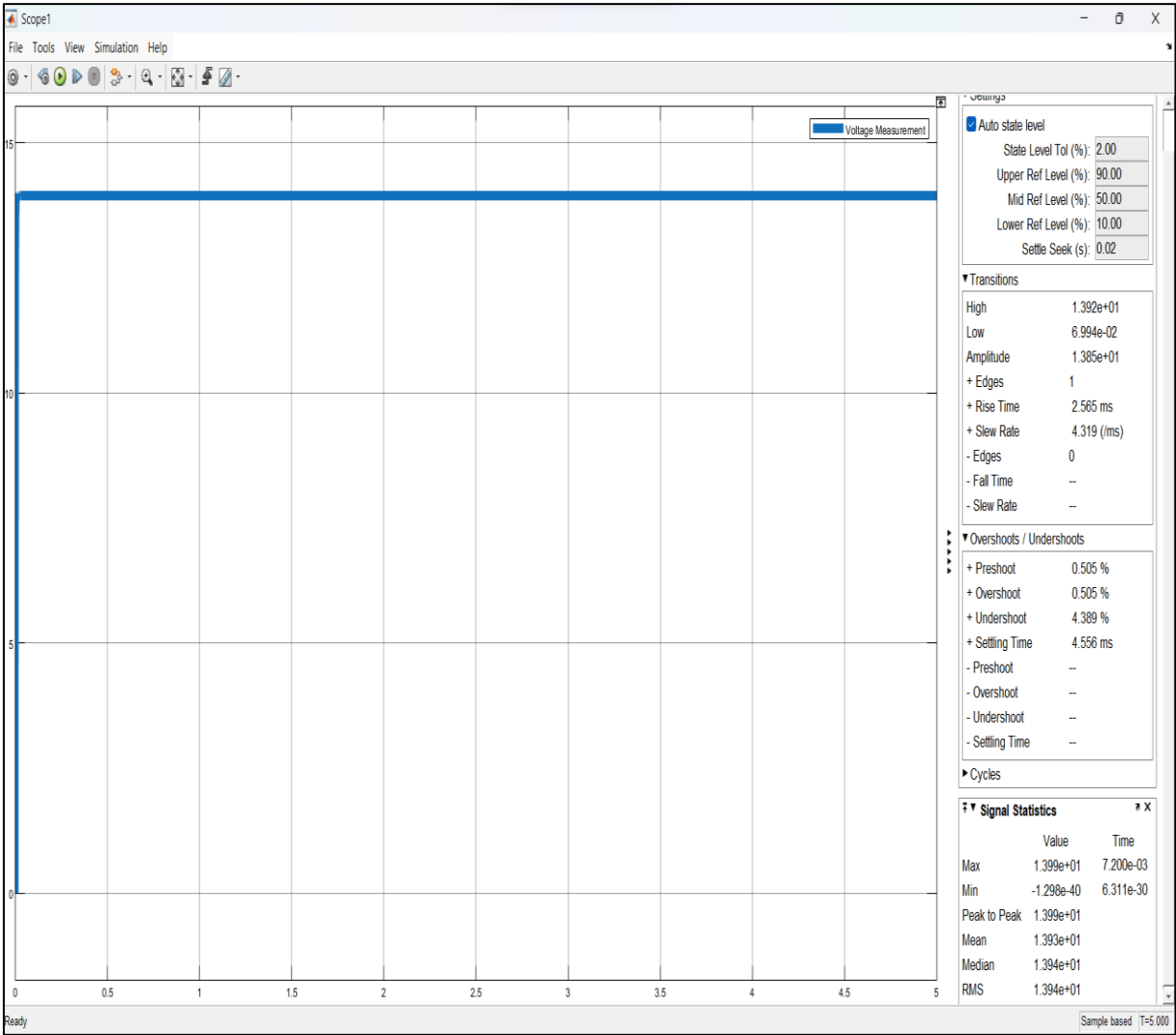


Fig 5.2: Time-Domain Output Voltage Response from MATLAB Simulation

The time-domain response shown in the above figure represents a fast and stable system behaviour. The output voltage rapidly rises from zero to its steady-state value of approximately 13.9 V with a very short rise time of about 2.565 ms, indicating a quick dynamic response. The system exhibits minimal overshoot (~0.5%) and negligible undershoot, demonstrating well-tuned control characteristics. Additionally, the settling time is around 4.556 ms, after which the signal remains nearly constant with no visible oscillations or fluctuations. This confirms that the system is highly stable, well-damped, and capable of maintaining a steady output under the given conditions, making it suitable for applications requiring precise and reliable voltage regulation.

Block Parameters of PID controller

Block Parameters: PID Controller

PID 1dof (mask) (link)

This block implements continuous- and discrete-time PID control algorithms and includes advanced features such as anti-windup, external reset, and signal tracking. You can tune the PID gains automatically using the 'Tune...' button (requires Simulink Control Design).

Controller: PID Form: Parallel

Time domain:

☒ Continuous-time

☐ Discrete-time

Discrete-time settings

Sample time (-1 for inherited): -1

Compensator formula

$$P + I \frac{1}{s} + D \frac{N}{1 + N \frac{1}{s}}$$

Main Initialization Saturation Data Types State Attributes

Controller parameters

Source: internal

Proportional (P): 1

Integral (I): 0 ☐ Use I*Ts (optimal for codegen)

Derivative (D): 0 ☐ Use externally sourced derivative

Filter coefficient (N): 100 ☒ Use filtered derivative

Automated tuning

Select tuning method: Transfer Function Based (PID Tuner App)

☒ Enable zero-crossing detection

Fig 5.3: Characteristics with Respect to Duty Cycles with a Fixed Solar Irradiation (Halogen Lamp)

The PID controller configuration shown in the figure is implemented in continuous-time using a parallel control structure. The controller is currently operating with only proportional action ($P = 1$), while the integral ($I = 0$) and derivative ($D = 0$) gains are disabled, effectively behaving as a proportional controller. This configuration provides a direct and immediate response to the error signal, ensuring fast system reaction. The derivative filter coefficient ($N = 100$) is defined to support future derivative action with noise attenuation, although it is not active in the present setup. The controller parameters are internally defined, and provision for automatic tuning using a transfer function–based approach (PID Tuner App) is available for further optimization. Overall, this configuration is suitable for achieving a stable and rapid response in systems where integral and derivative actions are not initially required, while still allowing flexibility for advanced tuning if needed.

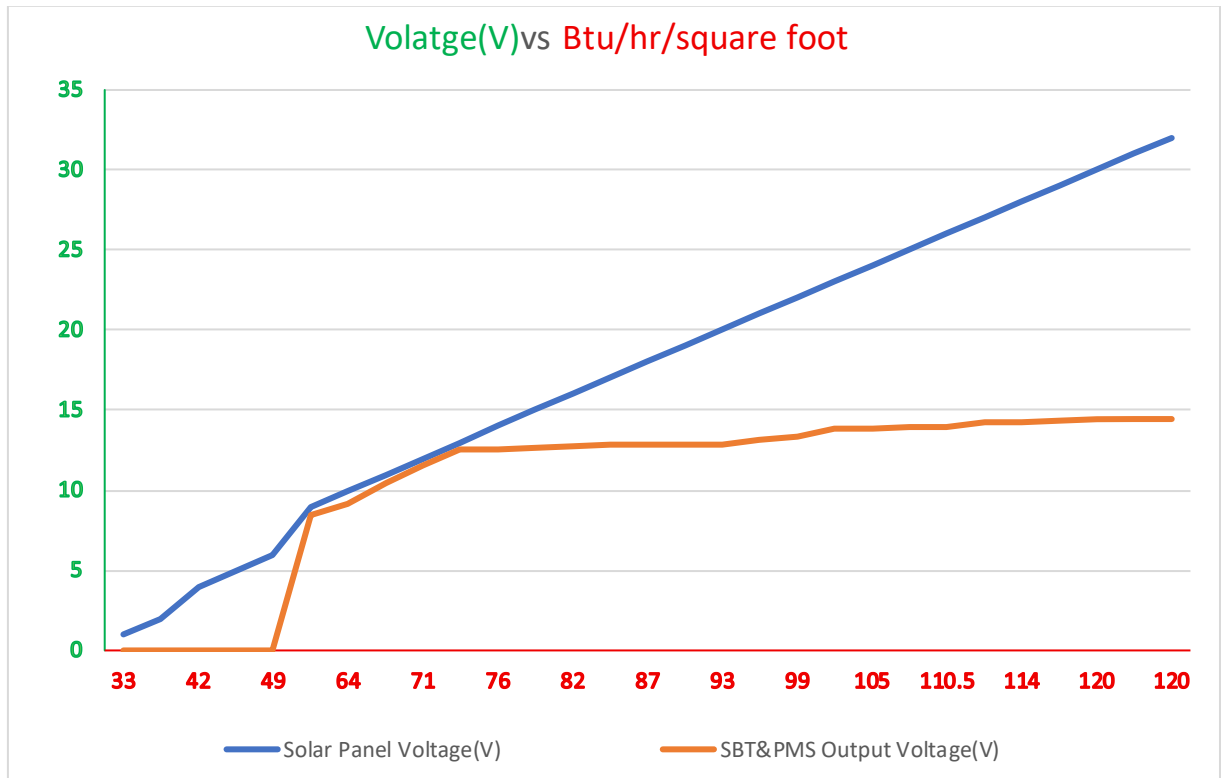


Fig 5.4: Voltage Profile of Solar Panel Voltage(V) and SBT&PMS Output Voltage(V) with respect to Btu/hr/square foot

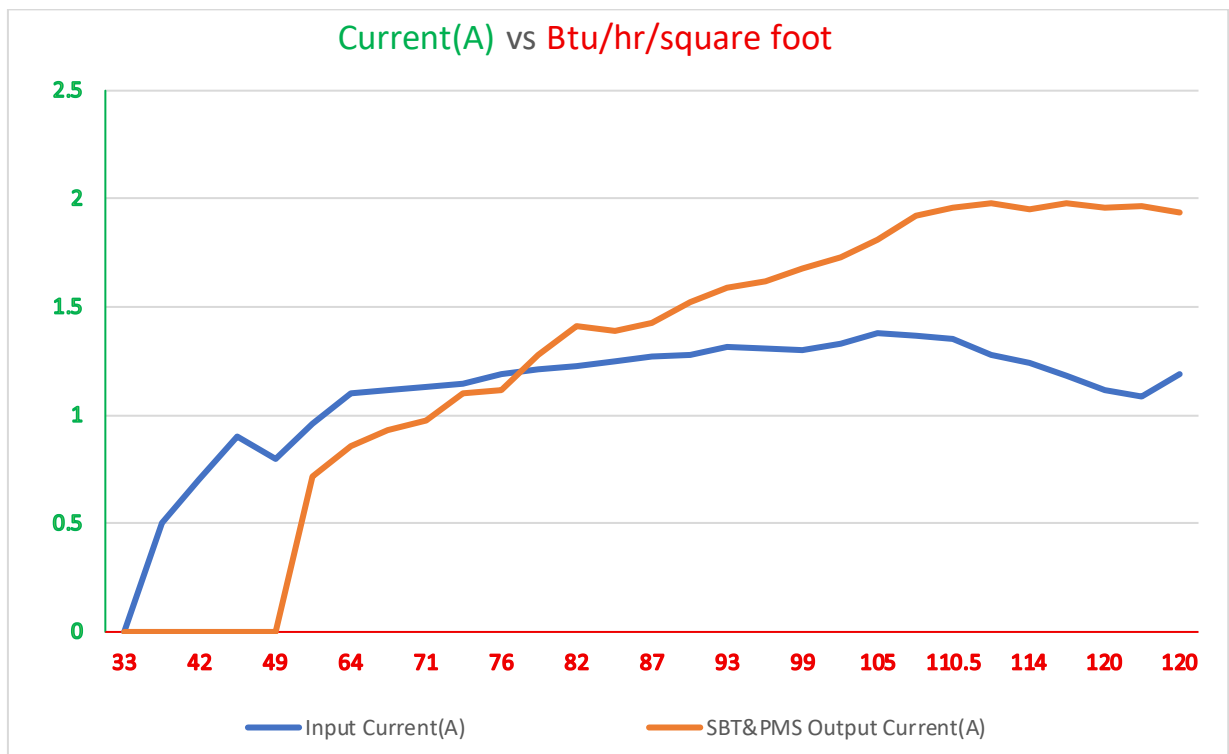


Fig 5.5: Current Profile of Solar Panel Current(A) and SBT&PMS Output Current(A) with respect to Btu/hr/square foot

Three-stage battery charging process used in Smart Battery Thermal and Power Management System for Photovoltaic Battery Charging:

1. Bulk Stage

2. Absorption Stage

3. Float Stage

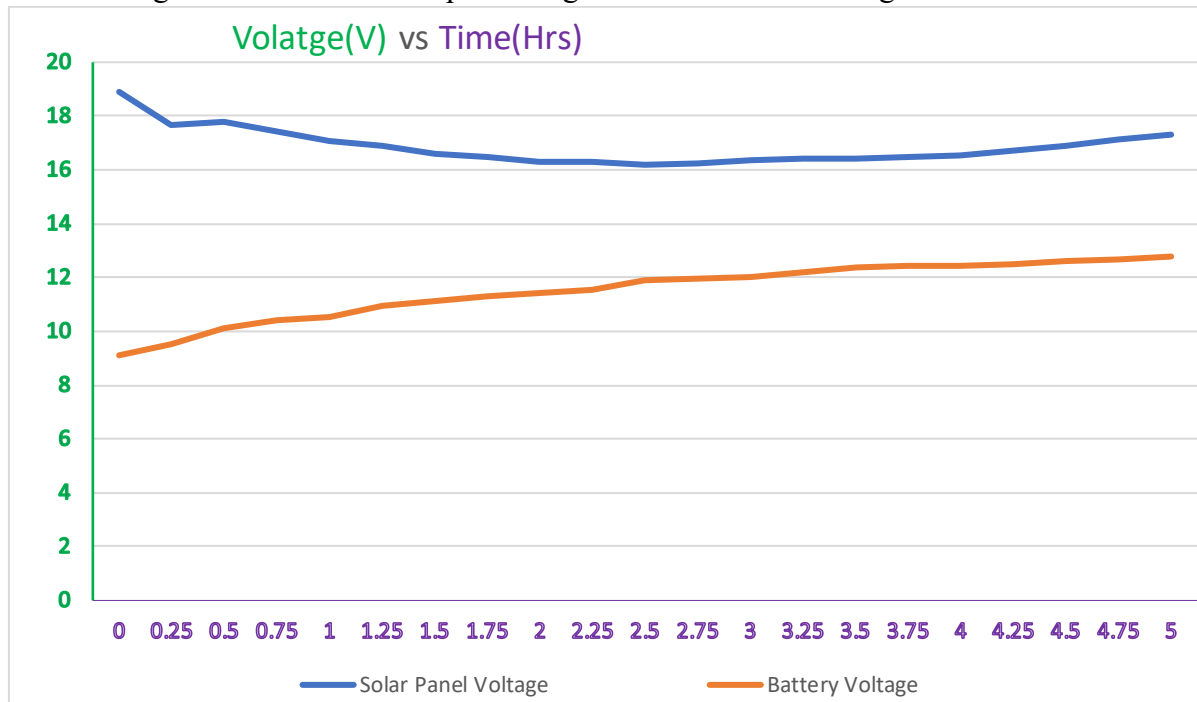


Fig 5.6: Voltage Profile of Solar Panel and 12v 6600mah Battery (3p Connection) with respective to Time 5hrs

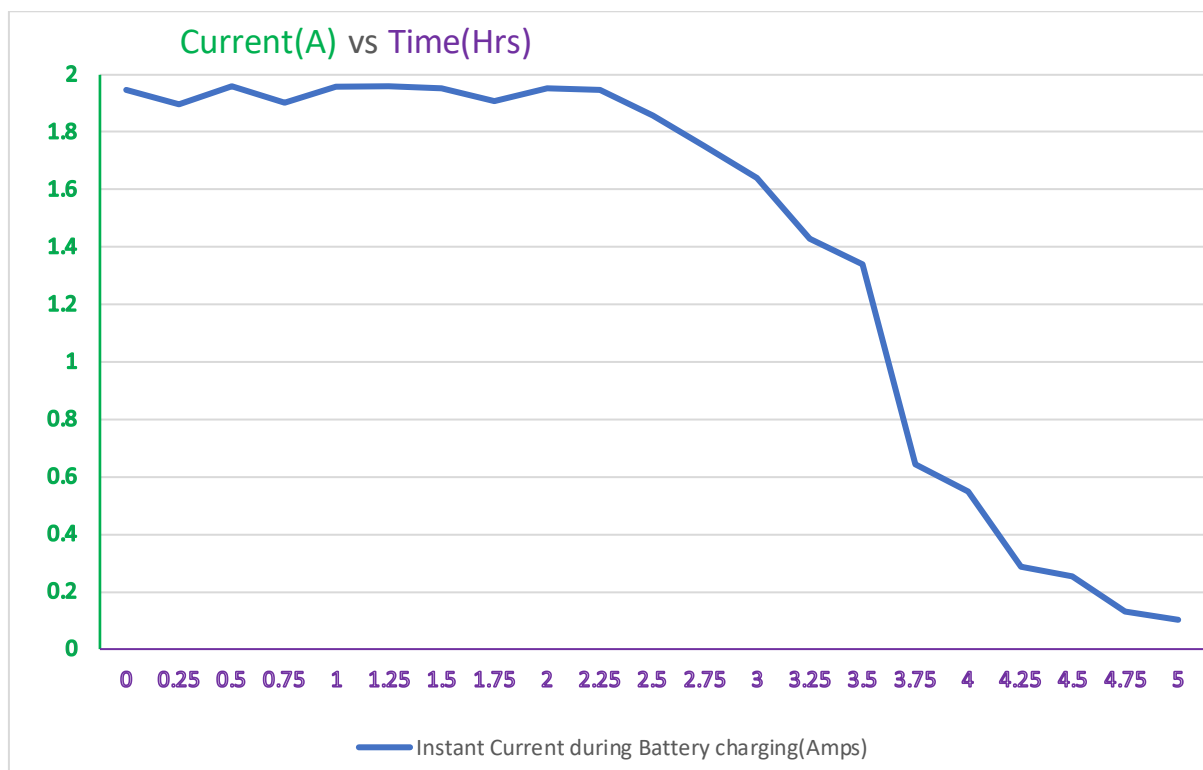


Fig 5.7: Current Profile of 12v 6600mah Battery with respective to Time

$$\text{SOC} = \frac{\text{Capacity Remaining}}{\text{Total Capacity}} * 100 * \text{hrs} \quad (14)$$

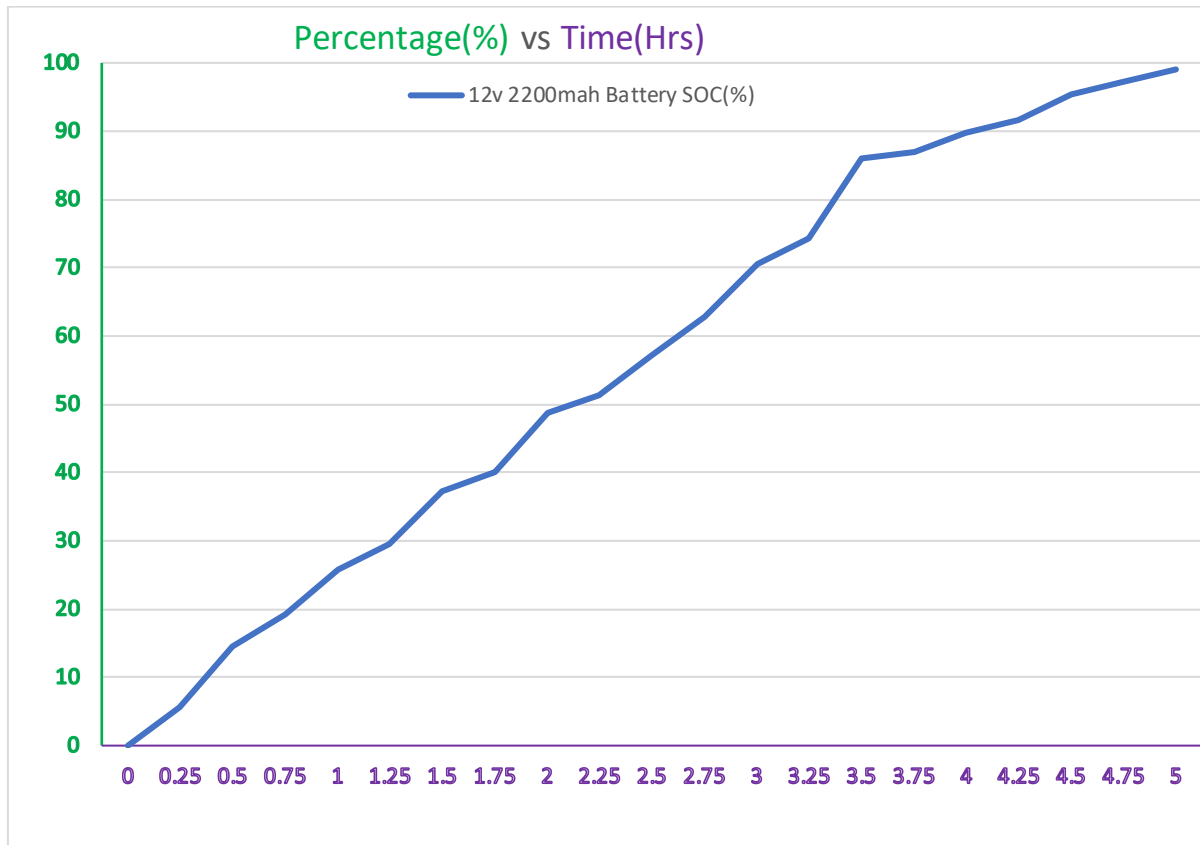


Fig 5.8: SOC (%) Profile of 12v 3*2200mah Battery (3cells in parallel) with respect to Time

Over Temperature Protection for Battery:

Table 7: Voltage Response to Battery Over temperature Condition

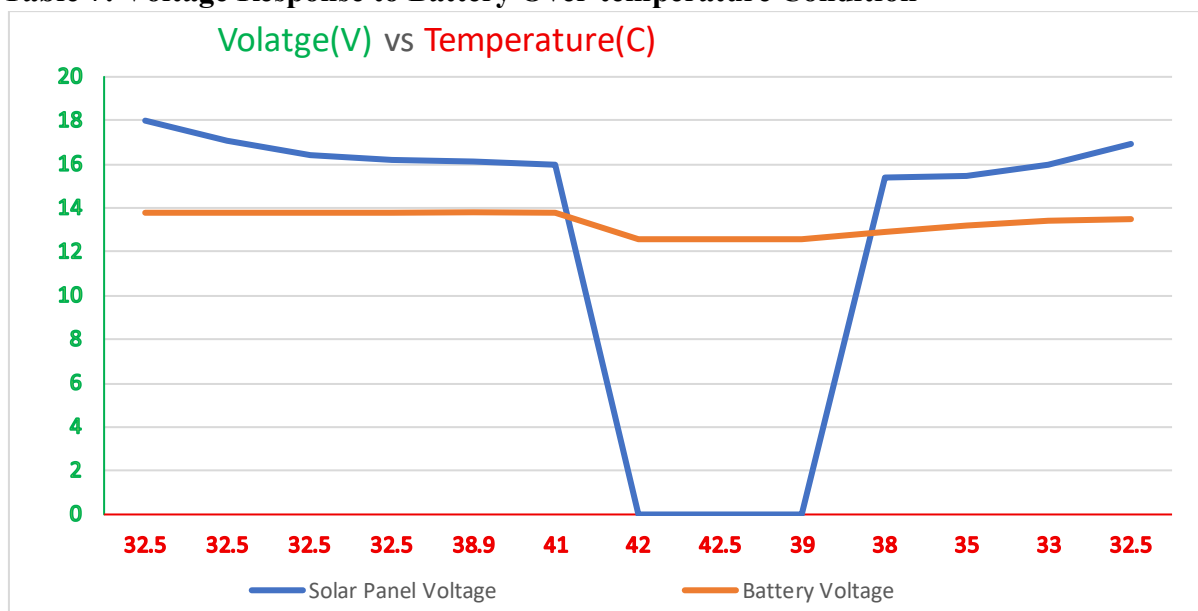


Fig 5.9: Voltage Response to Battery Overtemperature Condition

Table 5.1: Over Temperature Protection for AEM Electrolyzer:

Electrolyzer Temperature	Applied Voltage to Cartridge Heater (AC 230v)	Electrolyzer Voltage
67.5	232	2.1
68	232	2.2
67.8	232	2.2
68.5	232	2.1
69	232	2.2
70	0	2
70	0	0
70.9	0	0
69.8	0	0
68.3	0	0
66.9	232	0.1
64.6	232	2
66	232	2.2
65.8	232	2.3
66.1	232	2.4

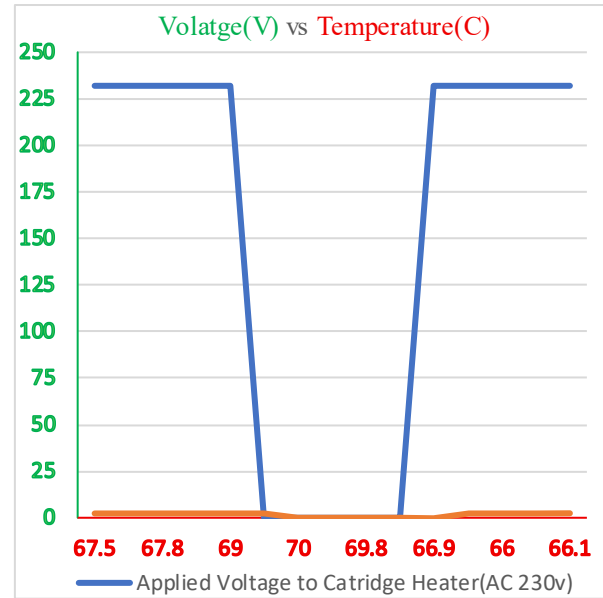
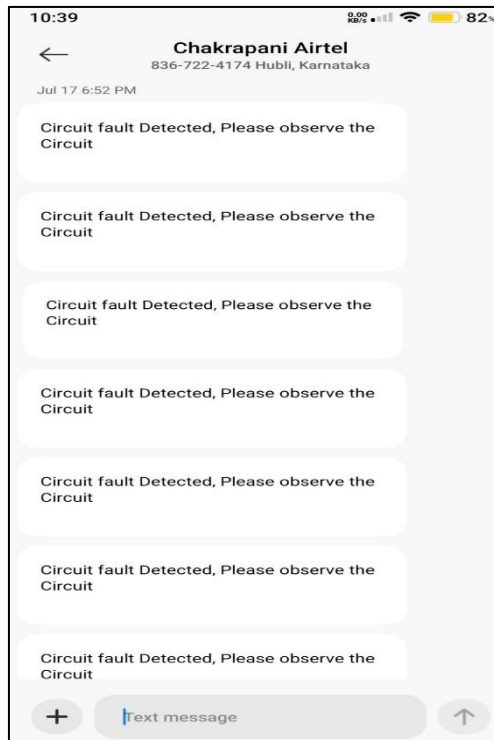


Fig 5.10: Voltage Response to AEM Electrolyzer Overtemperature Condition

Circuit Fault Detection and Alert System through SMS:

Fig 5.11: SMS-Based Alert Notification for Circuit Fault Detection



The SMS-Based Alert Notification for Circuit Fault Detection provides real-time supervision of the PV-Battery-Hydrogen Generation System and ensures rapid operator response during faults. During testing, the system successfully detected over-current conditions above 2.5 A flowing into battery during charging period, voltage deviations beyond $\pm 5\%$, and thermal rises above 45 °C, with SMS alerts delivered to humans to take direct action on the particular location. Safety mechanisms include hardware over-temperature cut-off when the electrolyzer exceeds 70 °C, resuming only after temperatures fall below 65 °C, preventing operation until stable conditions are restored.

The integrated fire/smoke sensor triggers LCD warnings, GSM (SIM900A) SMS notifications, and automatic system isolation during abnormal events, ensuring fast human intervention and enhancing overall operational safety and reliability.

5.2. AEM Electrolyzer Running Characteristics (V, I and P) for 5 min:

AEM electrolyzer was operated for five hours under controlled conditions. Throughout the test, when applied max voltage (V), achieved voltage & current at the electrolyzer, buck-converter output, and responses to power fluctuations were recorded.

Table 9: AEM Electrolyzer Running Characteristics (V, I and P) for 5min or 300sec

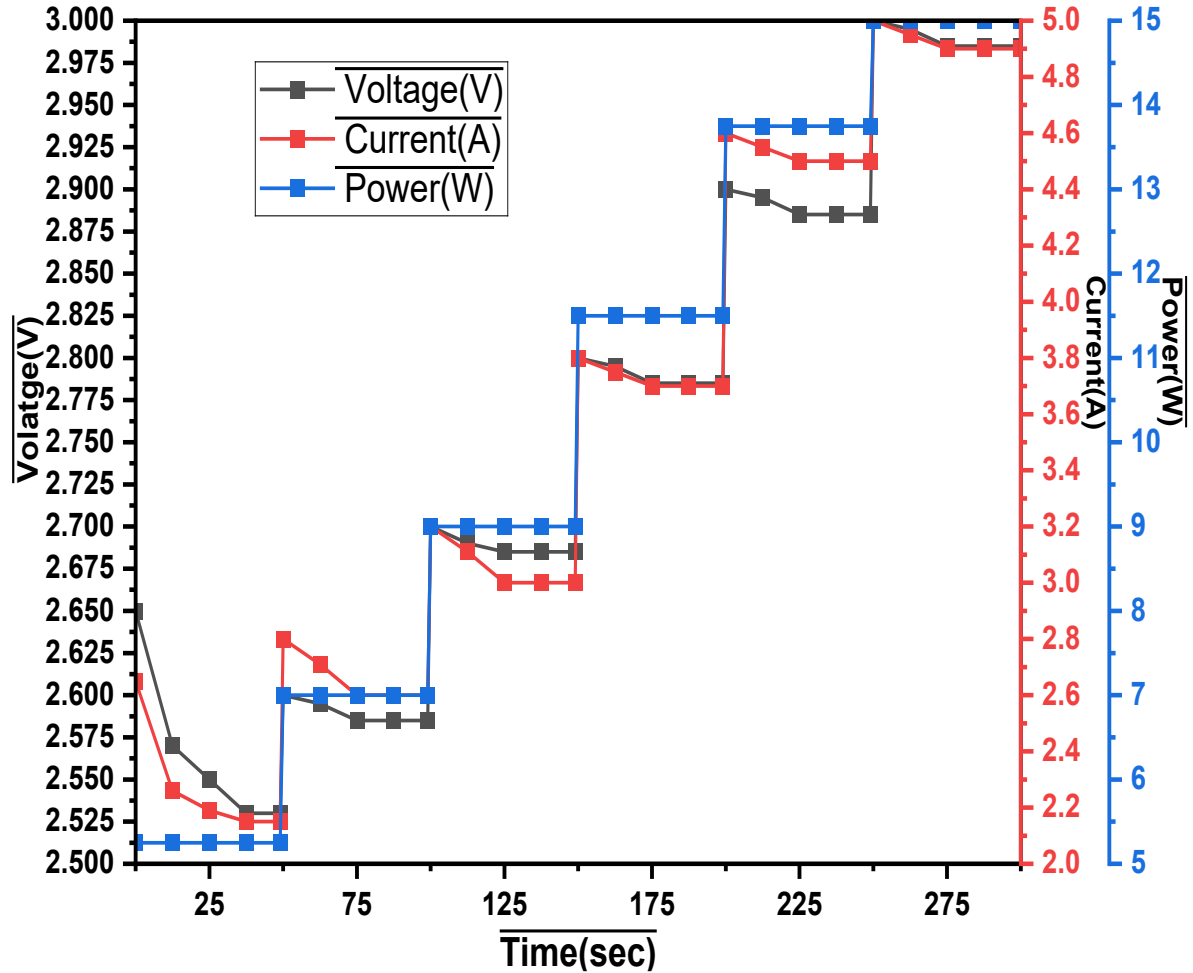


Fig 5.12: AEM Electrolyzer Running Characteristics for Voltage(V), Current(A), Power(W) with respective to Time(Sec)

Applied Voltage(V)	Achieved Voltage(V)
3V	2.985V
Applied Current(A)	Achieved Current(A)
5A	4.9A

5.3. Observation from Graph & table after Running the AEM Electrolyzer:

The plotted data illustrates how the AEM electrolyzer responds when voltage, current, and power are increased in a stepped manner over a 300-second run. Instead of using a linear rise, the experiment is carried out through 18 timed intervals where each parameter is held steady for a short period and then raised to the next level. As seen in the graph, voltage progresses from roughly 2.50 V to about 2.95 V, current moves from 2 A up to 5 A, and power correspondingly increases from around 5 W to nearly 15 W. Each flat segment represents a stable set point at which the electrolyzer is allowed to settle, while the sharp vertical transitions show the moments when the next step is applied. This pattern closely matches the actual behaviour observed during the running test, where the system is intentionally kept at fixed operating levels before shifting upward, helping to clearly identify how the electrolyzer stabilizes, how quickly it responds after each increment, and how the achieved values (2.985 V and 4.9 A) compare with the applied 3 V and 5 A set points.

5.4. Theoretical Calculation on Hydrogen Generated by AEM Electrolyzer Running Characteristics:

It is known that

Moles of Hydrogen per second(mol/s):

$$N_{H_2} = \frac{nI}{2F} \quad (15)$$

Hydrogen in Litres per minute(L/min):

$$V_{H_2} = N_{H_2}(\text{mol/s}) * 60 \quad (16)$$

Table 5.2 Theoretical Calculation on Hydrogen Generated by AEM Electrolyzer Running Characteristics

S.No.	Time (Sec)	Voltage (V)	Current (Amps)	Moles of Hydrogen per second, N_{H_2} (mol/s)	Hydrogen in Litres per minute, V_{H_2} (L/min)
1.	0-50Sec	2.53V	2.15A	0.0000111416 mol/s	0.0006685 L/min
2.	50-100Sec	2.585V	2.6A	0.00001347 mol/s	0.0008084 L/min
3.	100-150Sec	2.685V	3A	0.00001391 mol/s	0.0008348 L/min
4.	150-200Sec	2.785V	3.7A	0.00001937 mol/s	0.0011624 L/min
5.	200-250Sec	2.885V	4.5A	0.00002331 mol/s	0.001399 L/min
6	250-300Sec	2.985V	4.9A	0.00002539 mol/s	0.00152 L/min

Observation Based on the theoretical hydrogen-generation calculations and the recorded AEM electrolyzer running characteristics:

From the Fig 19 and Table 9, it is observed that the applied voltage–current steps produce reaction currents consistent with Faraday’s law, confirming that the electrolyzer generates hydrogen at the theoretically expected rate for each operating level. The smooth current stabilization, proportional rise in power, and absence of unstable spikes demonstrate efficient electrochemical behaviour, meaning the system converts electrical input into hydrogen output with predictable performance.

5.5 Full Cell Assembly and Chronoamperometry (CA) Analysis at 2.25 V using Electro Chemical Workstation

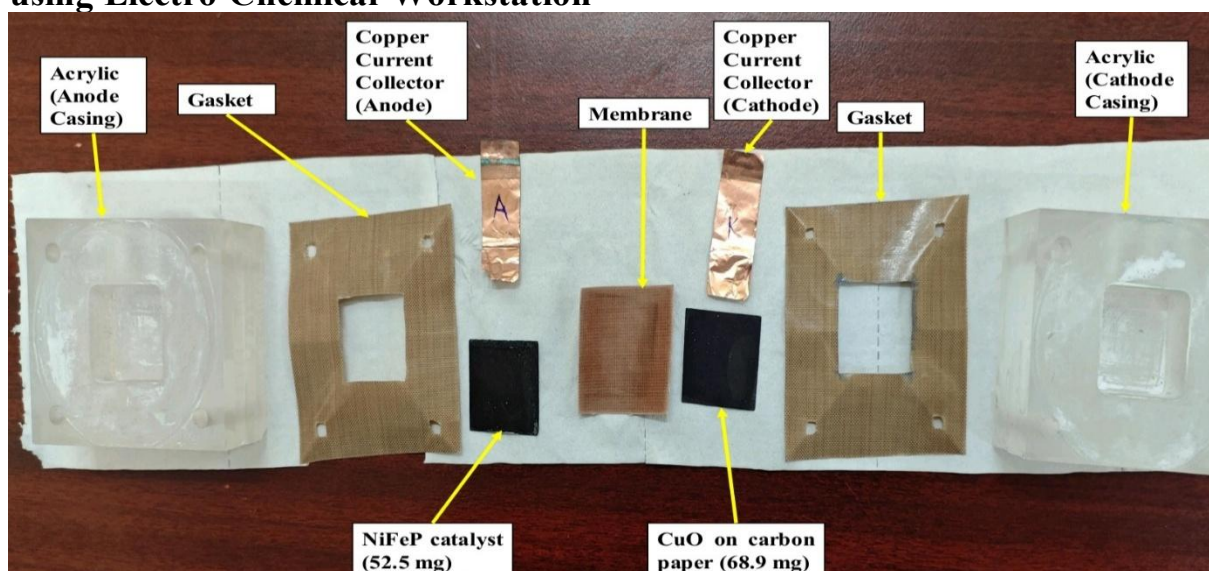


Fig 5.13: Full Cell Configuration and Electrode Assembly

Cell Type: Two-compartment acrylic electrochemical cell

Assembly sequence:

Acrylic → Gasket → Anode → Membrane → Cathode → Gasket → Acrylic

- Anode: NiFeP catalyst (52.5 mg)
- Function: Oxygen Evolution Reaction (OER)
- Cathode: NINC-IMI on carbon paper (68.9 mg)
- Catalyst loading: $\sim 5 \text{ mg/cm}^2$
- Membrane: Activated in 1 M KOH (12 hours)
- Enables OH^- ion transport
- Compact structure ensures efficient charge transfer
- Proper sealing prevents electrolyte leakage

Electro Chemical Workstation

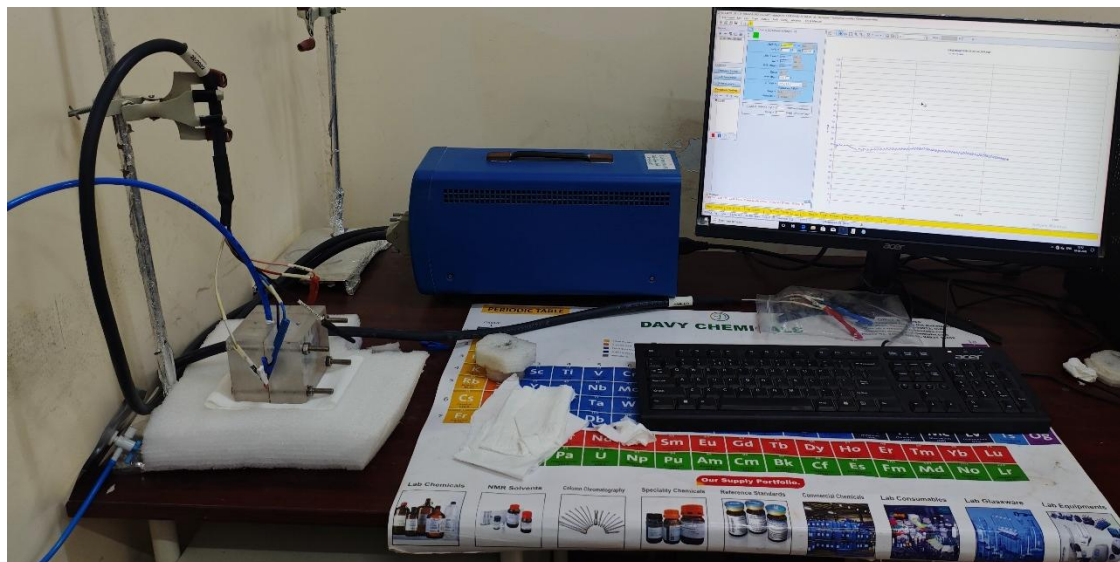
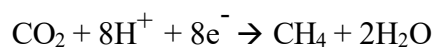


Fig 5.14: Chronoamperometry (CA) Analysis at 2.25 V using Electro Chemical Workstation

- CO₂ reduced at cathode under applied potential
- Multi-electron transfer reaction
- CA for 30 mins at 2.25 volts
- Formation of methane (CH₄) as target product
- Cathodic reaction:



- Anodic reaction:

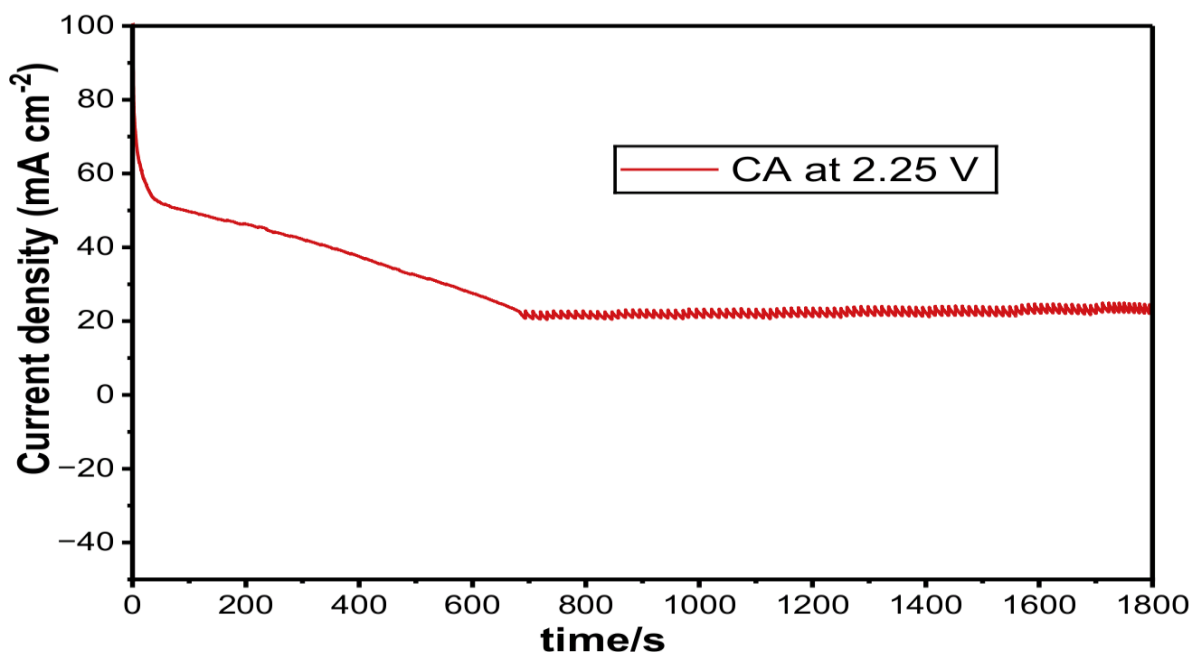
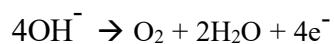


Fig 5.15: Chronoamperometric response of the system at an applied potential of 2.25 V showing current density variation with time.

CHAPTER 6

6.1 CONCLUSION

The developed Smart Battery Thermal and Power Management hardware system, integrating solar charging and fault detection mechanisms, highlights the potential of embedded technology in addressing critical energy management and safety challenges. The smart solar charger, with its combination of hardware sensors, real-time monitoring via OLED, and PID-based control, ensures efficient, safe, and autonomous battery charging even under fluctuating solar conditions. It incorporates protective features like temperature and voltage hysteresis, making it robust for renewable energy applications. Meanwhile, the Circuit Fault Detection and Alert System offer a practical safety solution, detecting hazardous gas leaks or circuit anomalies and immediately alerting users through both visual LCD feedback and GSM-based SMS notifications. Together, these works showcase how microcontroller-based systems can improve both operational efficiency and safety in diverse scenarios.

The systems have been enhanced with additional Real-time DS18B20 data monitors ambient, and electrolyzer temperatures, while voltage feedback ensures precise regulation. For the AEM electrolyzer (40 - 70 °C), a 50 W cartridge heater raises temperature to 40 °C within 5 - 10 min under insulation, with a microcontroller-driven solid-state relay in time-proportioning mode maintaining $\pm 0.5 - 1$ °C stability. And also, in this design integrated with automated cold-start capability, precise thermal control and regulation, efficient PV power utilization, and robust fault protection, thereby enhancing the operational efficiency of green hydrogen production and contributing to sustainable energy storage solutions for off-grid and renewable applications that prioritize efficiency, safety, and resilience.

6.2 Future Scope

The proposed system can be further enhanced by incorporating artificial intelligence and machine learning techniques to make it more adaptive and intelligent. Instead of relying only on conventional control methods, future work can explore data-driven approaches for predicting key battery parameters such as State of Charge (SoC) and State of Health (SoH). This would help in improving charging accuracy, extending battery life, and enabling early detection of abnormal operating conditions in renewable energy storage systems.

The system can also be extended by integrating IoT-based monitoring, allowing real-time tracking and control through cloud platforms or mobile applications. This would make the system more user-friendly and suitable for remote operation and large-scale deployments.

Future work includes the integration of artificial intelligence and machine learning techniques for predictive control, fault diagnosis, and energy optimization. Advanced models can be developed for accurate estimation of battery state parameters and solar power forecasting. The system can be extended with IoT-enabled remote monitoring, digital twin simulation, and intelligent energy management strategies. Furthermore, optimization algorithms and edge AI implementation can enhance real-time decision-making, efficiency, safety, and scalability of the PV–Battery–Hydrogen generation system.

Overall, these developments can transform the proposed system into a more efficient, reliable, and autonomous energy management solution, capable of adapting to dynamic conditions with minimal human intervention.

References

- [1] Kikuchi Y, Ichikawa T, Sugiyama M, Koyama M. Battery-assisted low-cost hydrogen production from solar energy: rational target setting for future technology systems. Published on 13 December 2018. At international journal of hydrogen energy 44 (2019) 1451 e1465. <https://doi.org/10.1016/j.ijhydene.2018.11.119>
- [2] Cai X, Lin R, Xu J, Lu Y. Construction and analysis of photovoltaic directly coupled conditions in PEM electrolyzer. Published on 24 December 2021. At Int J Hydrogen Energy Feb. 2022;47(10): 6494–507. <https://doi.org/10.1016/j.ijhydene.2021.12.017>
- [3] Alhaj Omar F. A new approach for improving the efficiency of the indirectly coupled photovoltaic-electrolyzer system. Published on 22 December 2022. At Int J Hydrogen Energy 48 (2023) 8768 e8782. <https://doi.org/10.1016/j.ijhydene.2022.11.327>
- [4] Yodwong B, Guilbert D, Phattanasak M, Kaewmanee W, Hinaje M, Vitale G. AC-DC converters for electrolyzer applications: state of the art and future challenges. Published on 29 May 2020. At *Electronics* **2020**, 9, 912; <https://doi.org/10.3390/electronics9060912>
- [5] Alonge F, Collura Stefania Maria, D'Ippolito Filippo, Guilbert D, Luna M, Vitale G. Design of a robust controller for DC/DC converter–electrolyzer systems supplied by μ wecss subject to highly fluctuating wind speed. Published on 10 March 2020. At Control Engineering Practice 98 (2020) 104383. <https://doi.org/10.1016/j.conengprac.2020.104383>
- [6] Mustafa Ergin Sahin. A photovoltaic powered electrolysis converter system with maximum power point tracking control. Published on 13 February 2020. At International Journal of Hydrogen Energy. <https://doi.org/10.1016/j.ijhydene.2020.01.162>
- [7] Awad M, Mahmoud MM, Elbarbary ZMS, Mohamed Ali L, Fahmy SN, Omar AI (2023) Design and analysis of photovoltaic/wind operations at MPPT for hydrogen production using a PEM electrolyzer: Towards innovations in green technology. Published on July 20, 2023. At plos ONE 18(7): e0287772. <https://doi.org/10.1371/journal.pone.0287772>
- [8] Raj Kapur Kumar, Paulson Samuel. Designing a hydrogen generation system through PEM water electrolysis with the capability to adjust fast fluctuations in photovoltaic power. Published on 29 July 2024. At International Journal of Hydrogen Energy 82 (2024) 1–10. <https://doi.org/10.1016/j.ijhydene.2024.07.376>

- [9] Galal Al-Muthanna, Shuhua Fang, Ibrahim AL-Wesabi, Khaled Ameer, Hossam Kotb, Kareem M. Aboras. Design and analysis of A High Speed MPPT Control Utilizing a Hybrid PSO-PID Controller under Partially Shaded Photovoltaic Battery Chargers. Published on 15 February 2023. At *Sustainability* **2023**, *15*, 3578.
<https://doi.org/10.3390/su15043578>
- [10] Bilal Taghezouit, Fouzi Harrou, Ying Sun and Walid Merrouche. Model-based fault detection in photovoltaic systems: A comprehensive review and avenues for enhancement. Published on 24 January 2024. At *Results in Engineering* **21** (2024) 101835. <https://doi.org/10.1016/j.rineng.2024.101835>.
- [11] Dhanraj J, Mostafaeipour A, Velmurugan K.; Techato K., Chaurasiya P.K., Solomon J.M., Gopalan A, Phoungthong K. An Effective Evaluation on Fault Detection in Solar Panels. Published on 19 November 2021. At *Energies* **2021**, *14*, 7770.
<https://doi.org/10.3390/en14227770>.
- [12] Suresh Muthusamy. Fault Detection and Monitoring of Solar PV Panels using Internet of Things. Published on 30.06.2018. At *International Journal of Industrial Engineering*, 2018;2(6):146-149. **ISSN: 2456-8449**.
<https://www.researchgate.net/publication/326913629>.
- [13] Peter Makeen, Hani A. Ghali and Saim Memon. Theoretical and Experimental Analysis of a New Intelligent Charging Controller for Off-Board Electric Vehicles Using PV Standalone System Represented by a Small-Scale Lithium-Ion Battery. Published on 16 June 2022. At *Sustainability* **2022**, *14*, 7396.
<https://doi.org/10.3390/su14127396>.
- [14] Caisheng Wang, M. Hashem Nehrir. Power Management of a Stand-Alone Wind/Photovoltaic/Fuel Cell Energy System. Published on 09.2008. At *IEEE TRANSACTIONS ON ENERGY CONVERSION*, VOL. 23, NO. 3, 2008.
<https://doi.org/10.1109/tec.2007.914200>
- [15] Fernando Valenciaga and Paul F. Puleston. Supervisor Control for a Stand-Alone Hybrid Generation System Using Wind and Photovoltaic Energy. Published on JUNE 2005. At *IEEE TRANSACTIONS ON ENERGY CONVERSION*, VOL. 20, NO. 2, 2005. <https://doi.org/10.1109/tec.2005.845524>
- [16] Tomonobu Senjyu, Toshiaki Nakaji, Katsumi Uezato, and Toshihisa Funabashi. A Hybrid Power System Using Alternative Energy Facilities in Isolated Island. Published on JUNE 2005. At *IEEE TRANSACTIONS ON ENERGY CONVERSION*, VOL. 20, NO. 2, 2005. <https://doi.org/10.1109/tec.2004.837275a>

- [17] Y. H. Li, S. S. Choi and S. Rajakaruna. An Analysis of the Control and Operation of a Solid Oxide Fuel-Cell Power Plant in an Isolated System. Published on JUNE 2005. At IEEE TRANSACTIONS ON ENERGY CONVERSION, VOL. 20, NO. 2, 2005. <https://doi.org/10.1109/tec.2005.847998>
- [18] Kodjo Agbossou Mohanlal Kolhe, Jean Hamelin, and Tapan K. Bose. Performance of a Stand-Alone Renewable Energy System Based on Energy Storage as Hydrogen. Published on SEPTEMBER 2004. At IEEE TRANSACTIONS ON ENERGY CONVERSION, VOL. 19, NO. 3, 2004. <https://doi.org/10.1109/tec.2004.827719>
- [19] Muhammad Bakr Abdelghany, Ahmed Al-Durra and Fei Gao. A Coordinated Optimal Operation of a Grid-Connected Wind-Solar Microgrid Incorporating Hybrid Energy Storage Management Systems. Published on JANUARY 2024. At IEEE TRANSACTIONS ON SUSTAINABLE ENERGY, VOL. 15, NO. 1, 2024. <https://doi.org/10.1109/tste.2023.3263540>
- [20] Abdulrahman M. Abomazid, Nader A. El-Taweel and Hany E. Z. Farag. Optimal Energy Management of Hydrogen Energy Facility Using Integrated Battery Energy Storage and Solar Photovoltaic Systems. Published on JULY 2022. At IEEE TRANSACTIONS ON SUSTAINABLE ENERGY, VOL. 13, NO. 3, 2022. <https://doi.org/10.1109/tste.2022.3161891>
- [21] Abdelfatah Ali, Mostafa F. Shaaban and Karar Mahmoud. Optimizing Hydrogen Systems and Demand Response for Enhanced Integration of RES and evs in Smart Grids. Published on MARCH 2025. At IEEE TRANSACTIONS ON SMART GRID, VOL. 16, NO. 2, 2025. <https://doi.org/10.1109/tsg.2024.3491774>

Publications by the candidate

1 Patents and Patent Applications

[1] Tatipamula Chakrapani¹, Kanasottu Anil Naik² and Leela Manohar Aeshala^{1*}.

"A System and Method for Intelligent Thermal and Power Management of Batteries in PV-Driven Hydrogen Production Systems" Indian Patent Outward Correspondence Application No. 421, filed March 17, 2026. (Patent Pending).

2 Conference Presentations

[2] Tatipamula Chakrapani and Leela Manohar Aeshala. "Design and Development of Smart Battery Thermal and Power Management System for PV-Battery-Hydrogen Generation System." Presented at the Global Conference for Decarbonization of Energy and Materials (GCDEM 2025), Singapore, October 16–19, 2025.

3 Awards and Recognitions

[3] Tatipamula Chakrapani, "Design and Development of Smart Battery Thermal Management System for Power Transfer from PV-Battery-Hydrogen Generation System." First Prize (Oral Presentation, PG Category) at the International e-Conference on Multifaceted Applications in Biotechnology and Applied Sciences (e-COMBAS 2.0), organized by EduLeviat Pvt. Ltd., August 29–31, 2025.

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